Real-Time Time-Dependent Density Functional Theory for Periodic and Spin-Polarized Systems with the Real-Space Multigrids (RMG)[†]

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Abstract

[Abstract: to be written last, after all sections are complete. Approximately 200 words. Should mirror paper 1 abstract structure: (1) what we present, (2) what problem it solves, (3) key methodological contribution, (4) validation/benchmarking, (5) scalability demonstration and outlook.]

1 Introduction

Real-time time-dependent density functional theory (RT-TDDFT) has established itself as one of the primary tools for simulating the electronic response of materials to external electromagnetic perturbations.[1,2] Its applications span a broad range of phenomena, including optical spectroscopy,[3–7] charge transfer and photovoltaics,[2,8,9] nanoplasmonics,[10–12] and spin and magnetization dynamics.[13–15] In an earlier publication, we presented a scalable RT-TDDFT implementation within the Real-space MultiGrid (`RMG`) code for finite molecular systems, demonstrating simulations of up to 24,000 electrons on massively parallel architectures.[?] The present work extends that implementation to periodic systems and spin-polarized calculations, significantly broadening the range of physically relevant problems accessible to `RMG`.

The transition from finite to periodic systems in RT-TDDFT requires a corresponding transition in the treatment of the electromagnetic perturbation. In finite systems, the length gauge—in which the coupling to an electric field is expressed through the dipole operator $\hat{\mathbf{r}}$—is natural and convenient. In periodic systems, however, the position operator is ill-defined because it is not translationally invariant, and the dipole moment of an extended system is intrinsically multivalued.[?,?] The correct framework for periodic systems is the velocity gauge, in which the electromagnetic interaction is expressed through the vector potential $\mathbf{A}(t)$ and the current operator, which is inherently translationally invariant. The transition between the two gauges is effected by a gauge transformation of the time-dependent Kohn-Sham equations,[?,?,?] and requires careful treatment of the nonlocal part of pseudopotentials.[?,?,?] The primary observable in the velocity gauge is the macroscopic current density $\mathbf{J}(t)$, from which the optical conductivity $\boldsymbol{\sigma}(\omega)$ and dielectric function $\boldsymbol{\varepsilon}(\omega)$ are extracted by Fourier analysis.

RT-TDDFT for periodic systems has been implemented in several codes. The `Octopus` code,[16] based on real-space grids, provides a comprehensive periodic RT-TDDFT implementation including both length and velocity gauges and has served as the primary validation reference for the present work. Plane-wave codes including `Qbox/Qball`,[17,18] `cp2k`,[19] and `Exciting`[?] offer periodic RT-TDDFT with natural handling of Born-von Kármán boundary conditions. Atom-centered basis set codes such as `SIESTA`[20] and `DFTB+`[21] provide efficient implementations for systems where localized orbitals are appropriate. The `DFT-FE` code[22] offers an alternative real-space approach based on finite elements. [Check: add any missing periodic RT-TDDFT codes before submission. In particular: GPAW, Abinit, Yambo.]

The treatment of spin-polarized systems within RT-TDDFT introduces an additional degree of freedom: the electronic density is resolved into spin-up and spin-down components, and the time-dependent exchange-correlation potential depends on both. In the collinear approximation, the two spin channels evolve independently subject to a common self-consistent spin-dependent potential, and the optical response carries spin-specific signatures. This capability is essential for magnetic insulators such as CrI_3, where the spin-up and spin-down optical spectra differ markedly and encode information about the electronic and magnetic structure.[13,15] The non-collinear spin extension, in which the electronic states are represented as two-component spinors, is beyond the scope of the present work and will be addressed in a future publication.

A distinctive feature of the `RMG` implementation is its scalability. The real-space multigrid approach assigns different regions of space to different processors, enabling near-linear scaling to hundreds or thousands of GPU-accelerated nodes.[23,?] This scalability is particularly significant for periodic calculations with spin, where both the k-point sampling and the spin-polarized density matrix increase the computational cost relative to finite systems. To demonstrate this capability, we apply the periodic spin-polarized RT-TDDFT implementation to nitrogen-vacancy (NV) centers in diamond—a prototypical point defect system of substantial interest for quantum sensing and quantum information applications.[?] By systematically increasing the supercell size from [X] to [Y] atoms, we show how the optical response of the defect converges as the interaction between periodic images becomes negligible, identifying the minimum supercell size required for physically meaningful results at realistic defect concentrations.

The remainder of this paper is organized as follows. Section 2 develops the theoretical framework, beginning with the interaction of matter with electromagnetic radiation in the long-wavelength approximation, proceeding through the gauge transformation from length to velocity gauge, the definition of the current operator including nonlocal pseudopotential contributions, the density matrix propagation scheme, the spin-collinear extension, and the postprocessing route from current to optical conductivity. Section 3 describes the algorithm as implemented in RMG and identifies the key approximations and their physical justification. Section 4 presents benchmarking calculations for one-dimensional (H_2 chain), two-dimensional (hexagonal BN), and three-dimensional (Si) periodic systems, followed by spin-polarized calculations for CrI_3, with comparisons to Octopus in all cases. The section concludes with the NV-center in diamond scalability demonstration. The paper concludes with a Summary and Outlook.

1.1 Interaction of Matter with Electromagnetic Radiation

The starting point for a RT-TDDFT calculation is the time-dependent Kohn–Sham equation for single-particle orbitals $\Psi_j(r,t)$ in the presence of external electromagnetic fields which, with the minimal coupling Hamiltonian, takes the form

$$\imath\frac{\partial}{\partial t}\Psi_j(\mathbf{r},t) = \left[\frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{q_e}{c}\mathbf{A}_{\text{ext}}(\mathbf{r},t)\right)^2 + q_e\Phi_{\text{ext}}(\mathbf{r},t) + \hat{V}_{\text{KS}}[\rho](t)\right]\Psi_j(\mathbf{r},t) \tag{1}$$

where $\mathbf{A}_{\text{ext}}$ and Φ_{ext} are vector and scalar potential representing external electromagnetic (EM) field, $\hat{\mathbf{p}} = -\imath\hbar\nabla$ is a canonical momentum operator, and $q_e = -1$ is and electron charge. The Kohn-Sham potential, $\hat{V}_{\text{KS}}[\rho](t)$, contains the ionic potential, Hartree potential, and exchange-correlation potential:

$$\hat{V}_{\text{KS}} = \hat{V}_{\text{ion}} + \hat{V}_{\text{H}}[\rho](\mathbf{r},t) + \hat{V}_{\text{xc}}[\rho](\mathbf{r},t). \tag{2}$$

In practical implementations based on norm-conserving or ultrasoft pseudopotentials, $\hat{V}_{\text{ion}}$ may contain both local and nonlocal parts, and gauge transformations of the Hamiltonian may introduce additional terms that must be accounted for in both the Hamiltonian and the current density operator. The choice of the specific form of $\mathbf{A}_{\text{ext}}$ and Φ_{ext} depends on the problem and is subject to a gauge freedom.[24]

For a monochromatic external EM field, the vector potential may be represented as a plane wave,

$$\mathbf{A}(r,t) = \mathbf{A}_0 e^{\imath(\mathbf{q}\cdot\mathbf{r}-\omega t)} + c.c. \tag{3}$$

where ω is the angular frequency, a wave vector $\mathbf{q}$ describes propagation direction, and $\mathbf{A}_0$ is a complex polarization amplitude. The corresponding electric and magnetic fields are obtained from Maxwell equations:

$$\begin{aligned}\mathbf{E}(\mathbf{r},t) =& -\nabla\Phi_{\text{ext}}(\mathbf{r},t) - \frac{1}{c}\frac{\partial\mathbf{A}_{\text{ext}}(\mathbf{r},t)}{\partial t},\\ \mathbf{B}(\mathbf{r},t) =& \nabla\times\mathbf{A}_{\text{ext}}(\mathbf{r},t).\end{aligned}$$

Any vector field $\mathbf{A}$ can be uniquely decomposed into a *transverse* (divergence-free) component perpendicular to $\mathbf{q}$ and a *longitudinal* (curl-free) component parallel to $\mathbf{q}$. In the Coulomb gauge, defined by the condition

$$\nabla\cdot\mathbf{A}_{\text{ext}}(\mathbf{r},t) = 0, \tag{4}$$

the vector potential is purely transverse: it carries only the transverse part of the electric field and the magnetic field, while the scalar potential Φ_{ext} carries the longitudinal part of the electric field exclusively. For a plane wave, this becomes apparent upon applying the gauge condition (4) to Eq. (3), which yields the transversality condition

$$\mathbf{q} \cdot \mathbf{A}_0 = 0, \tag{5}$$

requiring that the polarization of the radiation field is perpendicular to its propagation direction.

In general, the electromagnetic potentials ($\mathbf{A}$ and Φ) and the KS potential all appear simultaneously. However, the gauge freedom allows to transform $\mathbf{A}$ and Φ to form suitable for a given problem using an arbitrarily selected scalar function $\chi(r)$ and a set of tranformation rules

$$\mathbf{A} \to \mathbf{A} + \nabla\chi, \tag{6}$$

$$\phi \to \phi - \frac{1}{c}\frac{\partial\chi}{\partial t}, \tag{7}$$

$$\hat{U} = e^{-\imath\chi} = e^{\imath\mathbf{r}\cdot\mathbf{A}(t)} \tag{8}$$

$$\Psi \to \Psi' = \hat{U}\Psi. \tag{9}$$

The Hamiltonian transforms as:

$$\hat{H} \to \hat{H}' = \hat{U}\hat{H}\hat{U}^\dagger - \imath\hat{U}\frac{\partial\hat{U}^\dagger}{\partial t}. \tag{10}$$

A specific choice of χ relevant to periodic systems is discussed in Sec. **??**.

In chemistry applications, the most common choice for RT-TDDFT is the *length gauge*, in which $\mathbf{A} = 0$ and the external electromagnetic field enters entirely through a magnetic-field-free scalar potential. In the long-wavelength limit, applicable when the size of the system L satisfies $|\mathbf{q}|L \ll 1$ (i.e. L is much smaller than the radiation wavelength $\lambda = 2\pi/|\mathbf{q}|$), the external field is effectively spatially uniform over the system and the interaction reduces to the electric-dipole approximation. A spatially uniform electric field $\mathbf{E}(t)$ is represented by the dipolar scalar potential

$$\Phi(\mathbf{r}, t) = -\mathbf{E}(t) \cdot \mathbf{r}, \tag{11}$$

as can be verified from $\mathbf{E} = -\nabla\Phi$. This form is widely used to investigate the optical response of finite molecular systems.

While the plane-wave form (3) describes a monochromatic perturbation, in practice the time-oscillating factor may be replaced by an arbitrary envelope function $f(t)$, so that the applied electric field takes the form $\mathbf{E}(t) = \hat{\mathbf{e}}\, E_0\, f(t)$, where $\hat{\mathbf{e}}$ is the polarization direction. A particularly useful choice is the Dirac delta kick, $f(t) = \kappa\, \delta(t)$, which delivers a broadband impulsive perturbation of strength κ at $t = 0$. Since the delta function has a flat spectral content, a single time propagation yields the dynamical response at all frequencies simultaneously. The frequency-dependent absorption spectrum is then obtained from the Fourier transform of the time-dependent observable monitored during the propagation.

For finite systems, the relevant observable is the time-dependent dipole moment $\mathbf{d}(t) =$

$\int \mathbf{r}\rho(\mathbf{r},t)d\mathbf{r}^3$, from which the polarizability and absorption cross section are obtained from a dipole correlation function by Fourier transform. For periodic systems, however, the scalar-potential form $\Phi = -\mathbf{E}(t)\cdot\mathbf{r}$ is not compatible with periodic boundary conditions, because the position operator $\mathbf{r}$ is discontinuous at the cell boundary. This motivates the use of the velocity-gauge formulation, in which the external field enters through a spatially uniform vector potential $\mathbf{A}(t)$ and the scalar potential is set to zero. For finite systems, the optical response is extracted from the time-dependent dipole moment, which depends on the density, whereas for truly periodic systems the relevant observable is the current density $\mathbf{j}(\mathbf{r},t)$, which remains well defined under periodic boundary conditions.

In a summary, it is useful at this point to recognize that external perturbations in RT-TDDFT fall into two physically distinct categories. The first category comprises electromagnetic field perturbations, in which the external field enters through the vector potential in the kinetic energy via the minimal coupling $\hat{\mathbf{p}} \to \hat{\mathbf{p}} - (q_e/c)\mathbf{A}$. The choice of gauge (length or velocity) determines how this coupling is represented in the Hamiltonian, and the long-wavelength limit leads to the dipolar approximation discussed above. Typical applications include optical absorption spectra and laser-driven electron dynamics.

The second category comprises scalar potential perturbations of the form

$$\Phi_{\text{ext}}(\mathbf{r},t) = \Phi_0\, e^{\imath(\mathbf{q}\cdot\mathbf{r}-\omega t)} + \text{c.c.}, \tag{12}$$

which are added directly to the Kohn–Sham potential $\hat{V}_{\text{KS}}$ and involve no gauge freedom. The electric field generated by such a potential, $\mathbf{E} = -\nabla\Phi_{\text{ext}} \propto \imath\,\mathbf{q}\,e^{\imath(\mathbf{q}\cdot\mathbf{r}-\omega t)}$, is purely longitudinal (parallel to $\mathbf{q}$) and carries no magnetic field. This type of perturbation probes the density–density response function $\chi(\mathbf{q},\omega)$ and is the natural tool for studying collective excitations at finite momentum transfer, such as plasmon dispersion and charge-density oscillations.

Although both categories may involve the same spatial factor $e^{\imath\mathbf{q}\cdot\mathbf{r}}$, they are physically and computationally distinct: the electromagnetic perturbation couples to the electron momentum through the kinetic energy and requires gauge treatment, while the scalar perturbation couples to the electron density through the potential energy and enters the calculation without any gauge considerations.

Many important spectroscopies and collective excitations involve finite momentum transfer and lie beyond the dipolar approximation. These include plasmon dispersion $\varepsilon(\mathbf{q},\omega)$, electron energy-loss spectroscopy (EELS) where the beam transfers finite $\mathbf{q}$ to the sample, inelastic X-ray scattering (IXS), and magnon dispersion in magnetic systems. Table 1 summarizes the common external-field representations used in TDDFT and their typical applications.

1.2 VELOCITY GAUGE

For finite systems, the relevant observable is the time-dependent dipole moment $\mathbf{d}(t) = \int \mathbf{r}\,\rho(\mathbf{r},t)\,d\mathbf{r}$, from which the polarizability and absorption cross section follow by Fourier transform. For periodic systems, however, the scalar-potential form $\Phi = -\mathbf{E}(t)\cdot\mathbf{r}$ is not compatible with periodic boundary conditions, because the position operator $\mathbf{r}$ is discontinuous at the cell boundary. This motivates the use of the velocity-gauge formulation, in which the external field enters through a spatially uniform vector potential $\mathbf{A}(t)$ and the scalar

Table 1: Summary of common external-field representations used in TDDFT, applicable to both linear-response and real-time formulations. In Coulomb gauge, the scalar potential carries the longitudinal part of the electric field, while the transverse vector potential carries the transverse electric field and the magnetic field.

Case	Potentials	Field character	B field	Typical applications
Length gauge, dipole	$\Phi = -\mathbf{r}\cdot\mathbf{E}(t)$, $\mathbf{A} = 0$	homogeneous electric field	no	optical absorption of finite systems, photochemistry[?, ?]
Velocity gauge	$\Phi = 0$, $\mathbf{A} = \mathbf{A}(t)$	homogeneous field	no	optical response of solids and periodic systems under PBC[?, ?, ?]
Finite-$\mathbf{q}$ scalar probe	$\Phi \propto e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}$, $\mathbf{A} = 0$	longitudinal, $\mathbf{E} \parallel \mathbf{q}$	no	density response, loss function, EELS, plasmon dispersion[?, ?, ?]
Finite-$\mathbf{q}$ EM wave	$\mathbf{A} \propto \boldsymbol{\epsilon}\, e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}$, $\mathbf{q}\cdot\boldsymbol{\epsilon} = 0$	transverse EM radiation	yes	IXS, radiative coupling beyond dipole, magnon dispersion[?, ?]

potential is set to zero.

The time-dependent Kohn–Sham equation then takes the form

$$\imath \frac{\partial}{\partial t}\Psi_j(\mathbf{r},t) = \left[\frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{q_e}{c}\mathbf{A}(t)\right)^2 + \hat{V}_{\mathrm{KS}}[\rho](t)\right]\Psi_j(\mathbf{r},t), \tag{13}$$

where the two gauges are related by the Göppert-Mayer transformation, discussed in detail in Sec. **??**. In the velocity gauge, the position operator no longer appears in the Hamiltonian and all terms are compatible with periodic boundary conditions. Correspondingly, the physical observable used to extract the optical response is the time-dependent current density $\mathbf{j}(\mathbf{r},t)$ rather than the dipole moment, since the current is well defined under periodic boundary conditions whereas $\mathbf{d}(t)$ is not.

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1.3 REMAINING STUFF

To describe the interaction of electrons with an external electromagnetic field, we start from the time-dependent Kohn–Sham (TDKS) equation for the occupied Kohn–Sham orbitals,

$$i\frac{\partial}{\partial t}\psi_n(\mathbf{r},t) = \hat{H}_{\mathrm{KS}}(t)\psi_n(\mathbf{r},t), \tag{14}$$

where, in atomic units, the TDKS Hamiltonian in the presence of external scalar and vector potentials is written as

$$\hat{H}_{\mathrm{KS}}(t) = \frac{1}{2}\left[\hat{\mathbf{p}} + \frac{1}{c}\mathbf{A}_{\mathrm{ext}}(\mathbf{r},t)\right]^2 - \Phi_{\mathrm{ext}}(\mathbf{r},t) + \hat{V}_{\mathrm{ion}} + \hat{V}_{\mathrm{H}}[n](\mathbf{r},t) + \hat{V}_{\mathrm{xc}}[n](\mathbf{r},t). \tag{15}$$

Here $\hat{\mathbf{p}} = -i\nabla$, $\Phi_{\rm ext}(\mathbf{r},t)$ and $\mathbf{A}_{\rm ext}(\mathbf{r},t)$ are the external scalar and vector potentials, and $\hat{V}_{\rm ion}$, $\hat{V}_{\rm H}[n]$, and $\hat{V}_{\rm xc}[n]$ denote the ionic, Hartree, and exchange-correlation contributions, respectively. In practical implementations based on pseudopotentials, $\hat{V}_{\rm ion}$ may contain both local and nonlocal parts. The corresponding electric and magnetic fields are obtained from

$$\mathbf{E}(\mathbf{r},t) = -\nabla\Phi_{\rm ext}(\mathbf{r},t) - \frac{1}{c}\frac{\partial \mathbf{A}_{\rm ext}(\mathbf{r},t)}{\partial t}, \qquad \mathbf{B}(\mathbf{r},t) = \nabla\times\mathbf{A}_{\rm ext}(\mathbf{r},t). \tag{16}$$

For finite systems, such as atoms and molecules, the interaction with light is commonly treated within the long-wavelength, or electric-dipole, approximation. In this limit the spatial phase factor of the electromagnetic wave,

$$e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}, \tag{17}$$

varies negligibly over the extent of the system, so that $\mathbf{q}\cdot\mathbf{r} \ll 1$ and the field may be regarded as spatially uniform. The interaction is then conveniently written in the length gauge as

$$\hat{V}_{\rm ext}^{(L)}(t) = -\mathbf{r}\cdot\mathbf{E}(t). \tag{18}$$

Equivalently, one may regard this perturbation as arising from the scalar potential

$$\Phi_{\rm ext}^{(L)}(\mathbf{r},t) = \mathbf{r}\cdot\mathbf{E}(t), \qquad \mathbf{A}_{\rm ext}^{(L)}(\mathbf{r},t) = 0. \tag{19}$$

Within this approximation, the magnetic component of the radiation field is neglected, and the physically relevant quantity is the field polarization direction. Since the wavevector $\mathbf{q}$ is no longer retained explicitly, the terms transverse and longitudinal are not the most useful descriptors at this stage.

The length-gauge form of Eq. (18) is well suited for finite systems, but becomes problematic for periodic solids. In particular, the scalar potential $-\mathbf{E}(t)\cdot\mathbf{r}$ is not lattice periodic and is therefore incompatible with Bloch-periodic boundary conditions. In real-space representations this issue is often manifested as a discontinuous sawtooth-like potential across the cell boundary. More fundamentally, the response of a periodic solid to a homogeneous electric field is more naturally expressed in terms of macroscopic current or polarization rather than a direct scalar-potential coupling through the position operator.

For this reason, periodic RT-TDDFT is commonly formulated in the velocity gauge. For a spatially homogeneous electric field, a gauge transformation removes the scalar potential and introduces a purely time-dependent vector potential,

$$\Phi_{\rm ext}^{(V)}(\mathbf{r},t) = 0, \qquad \mathbf{A}_{\rm ext}^{(V)}(\mathbf{r},t) = \mathbf{A}(t), \qquad \mathbf{E}(t) = -\frac{1}{c}\frac{d\mathbf{A}(t)}{dt}. \tag{20}$$

The TDKS Hamiltonian then takes the form

$$\hat{H}_{\rm KS}^{(V)}(t) = \frac{1}{2}\left[\hat{\mathbf{p}} + \frac{1}{c}\mathbf{A}(t)\right]^2 + \hat{V}_{\rm ion} + \hat{V}_{\rm H}[n](\mathbf{r},t) + \hat{V}_{\rm xc}[n](\mathbf{r},t). \tag{21}$$

This formulation is gauge-equivalent to the dipole length gauge of Eq. (18), but is considerably better suited for periodic systems because the perturbation preserves translational symmetry. It is important to emphasize that the homogeneous-field velocity gauge of Eq. (20) is still a dipole-limit description. Since $\mathbf{A}(t)$ carries no spatial dependence, it does not represent a full propagating electromagnetic wave and does not generate a magnetic field, $\nabla \times \mathbf{A}(t) = 0$.

A more general description of radiative coupling is obtained by retaining the spatial dependence of the external field. In Coulomb gauge,

$$\nabla \cdot \mathbf{A}_{\text{ext}}(\mathbf{r}, t) = 0, \tag{22}$$

a transverse electromagnetic plane wave may be written as

$$\mathbf{A}_{\text{ext}}(\mathbf{r}, t) = \mathbf{A}_0 e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)} + \text{c.c.}, \qquad \mathbf{q} \cdot \mathbf{A}_0 = 0. \tag{23}$$

The condition $\mathbf{q} \cdot \mathbf{A}_0 = 0$ expresses the transverse character of the field, namely that the polarization direction is perpendicular to the propagation direction $\mathbf{q}$. In this case both electric and magnetic components are present, with

$$\mathbf{E}(\mathbf{r}, t) = -\frac{1}{c}\frac{\partial \mathbf{A}_{\text{ext}}(\mathbf{r}, t)}{\partial t}, \qquad \mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}_{\text{ext}}(\mathbf{r}, t). \tag{24}$$

This finite-$\mathbf{q}$ formulation is more general than the dipole approximation because it explicitly retains momentum transfer and spatial phase information. In the long-wavelength limit, $e^{i\mathbf{q}\cdot\mathbf{r}} \approx 1$, it reduces to the homogeneous-field description discussed above.

It is useful to distinguish the transverse plane wave of Eq. (23) from a scalar-potential perturbation of the form

$$\Phi_{\text{ext}}(\mathbf{r}, t) = \Phi_0\, e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}, \qquad \mathbf{A}_{\text{ext}}(\mathbf{r}, t) = 0. \tag{25}$$

In this case the electric field is

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \Phi_{\text{ext}}(\mathbf{r}, t) = -i\mathbf{q}\, \Phi_0\, e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}, \tag{26}$$

which is parallel to $\mathbf{q}$. Such a perturbation is therefore longitudinal and probes the density response of the system at finite momentum transfer. This type of field is natural for the calculation of density-density response functions, loss spectra, and plasmonic excitations, whereas the transverse vector-potential form of Eq. (23) describes radiative coupling to an external electromagnetic wave.

The distinction between these external-field representations is also reflected in the analysis of RT-TDDFT simulations. For finite systems treated in the dipole approximation, the response is commonly characterized through the time-dependent dipole moment and its Fourier transform, which yield optical absorption spectra and frequency-dependent polarizabilities. In periodic systems formulated in the velocity gauge, the natural observable is instead the time-dependent current density, from which optical conductivity, dielectric response, and related quantities may be extracted. In the exact limit, length-gauge and velocity-gauge descriptions are equivalent, provided that observables and operators are transformed consistently. In practice, however, the current-based formulation is the natural choice for periodic systems under homogeneous fields, while the more general finite-$\mathbf{q}$ framework provides a clear route toward momentum-resolved response beyond the dipole limit.

In the present work we employ the homogeneous-field velocity-gauge formulation of Eqs. (20) and (21) as the working framework for periodic RT-TDDFT. This choice preserves periodic boundary conditions, provides direct access to current-based observables, and establishes a natural starting point for future extensions to finite-$\mathbf{q}$ perturbations and momentum-resolved spectroscopies.

2 Methodology

All equations in this section are given in atomic units ($\hbar = m_e = e = 4\pi\varepsilon_0 = 1$) unless stated otherwise.

2.1 Overview of RMG

[Brief overview of RMG: real-space grid, multigrid acceleration, GPU scalability. Refer heavily to paper 1[?] and the RMG code paper[23] rather than repeating. Keep to 1–2 paragraphs. Focus on what is new or relevant for the periodic case: k-point sampling, Bloch states, periodic boundary conditions. Mention norm-conserving pseudopotentials (NCPP only).]

[Note from CONTEXT.md: the code uses norm-conserving PP only (asserted in RmgTddft.cpp line 287). Ultrasoft PP requires S≠1 in the propagation (eldyn_nonort.cpp path) and is not yet implemented. State this as an explicit scope limitation here.]

2.2 Interaction of Matter with Electromagnetic Radiation

[DERIVATION NEEDED from Jacek's LaTeX files. Content outline: - Start from the full minimal-coupling Hamiltonian with EM field treated as a plane wave: $\exp(\imath\omega t - \imath\mathbf{q}\cdot\mathbf{r})$ - Derive the long-wavelength (dipole) approximation: $\mathbf{q} \to 0$, i.e., $\lambda \gg$ system size, valid for optical/UV and unit-cell scales - Show that in this limit the vector potential $\mathbf{A}$ becomes spatially uniform: $\mathbf{A}(\mathbf{r}, t) \to \mathbf{A}(t)$ - Result: the two natural gauge choices in the long-wavelength limit are the length gauge (scalar potential $\Phi = -\mathbf{r}\cdot\mathbf{E}$, $\mathbf{A} = 0$) and the velocity gauge ($\Phi = 0$, $\mathbf{A}(t)$ uniform) - Physical interpretation: length gauge couples to position (dipole), velocity gauge couples to momentum (current) - Reference: Mattiat & Luber (2022) Section 2.3; Yabana & Bertsch (1996)]

The time-dependent Kohn-Sham (TDKS) equations governing the electronic dynamics under an external electromagnetic perturbation take the form

$$\imath\frac{\partial}{\partial t}\psi_n(\mathbf{r}, t) = \hat{H}_{\mathrm{KS}}(t)\,\psi_n(\mathbf{r}, t), \tag{27}$$

where $\hat{H}_{\mathrm{KS}}(t)$ is the time-dependent Kohn-Sham Hamiltonian. [Expand: write out $\hat{H}_{\mathrm{KS}}(t)$ with all terms including the EM coupling, then derive the long-wavelength limit.]

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2.3 Gauge Transformation: Length to Velocity Gauge

Content outline: - State clearly why length gauge fails for periodic systems: the position operator $\hat{\mathbf{r}}$ is not translationally invariant; dipole moment of a periodic system is multivalued (refer to modern theory of polarization: Resta 1994, King-Smith & Vanderbilt 1993) - Present the Göppert-Mayer gauge transformation explicitly: gauge function $\chi(\mathbf{r}, t) = \mathbf{r}\cdot\mathbf{A}(t)$, transformations of potentials and wavefunctions (Mattiat eqs. 15–19) - Show that the length gauge TDKS (eq. 28) transforms to the velocity gauge TDKS (eq. 29) - CRITICAL: the nonlocal pseudopotential Vˆnl does NOT commute with the gauge transformation. The gauge-transformed Vˆnl acquires an additional term (Mattiat eq. 23). Derive this explicitly here or refer to Appendix A. - Result: velocity gauge TDKS Hamiltonian with A(t) coupling - Reference: Mattiat & Luber (2022) Sections 2.3.1, Appendix A, B; Pickard & Mauri (2001)

In the length gauge, the TDKS equations read

$$\imath \frac{\partial}{\partial t}\psi_n(\mathbf{r},t) = \left[\hat{H}_0 + \mathbf{r}\cdot\mathbf{E}(t)\right]\psi_n(\mathbf{r},t), \tag{28}$$

where $\hat{H}_0$ is the unperturbed Kohn-Sham Hamiltonian and $\mathbf{E}(t)$ is the electric field. Applying the gauge transformation $\chi(\mathbf{r},t) = \mathbf{r}\cdot\mathbf{A}(t)$, where the vector potential satisfies $\mathbf{E}(t) = -\partial\mathbf{A}/\partial t$, the velocity gauge TDKS equations take the form

$$\imath \frac{\partial}{\partial t}\psi_n(\mathbf{r},t) = \left[\frac{1}{2}\left(\hat{\mathbf{p}} + \mathbf{A}(t)\right)^2 + V(\mathbf{r},t) + \tilde{V}^{\mathrm{nl}}(\mathbf{r},t)\right]\psi_n(\mathbf{r},t), \tag{29}$$

where $\tilde{V}^{\mathrm{nl}}$ denotes the gauge-transformed nonlocal pseudopotential.[?, ?] [Expand: write out all terms of $\tilde{V}^{\mathrm{nl}}$ explicitly. Derive the linearization in A(t) that gives the perturbative coupling term. Connect to Appendix A.]

2.4 Vector Potential and Magnetic Perturbations

Content outline: - Show that the velocity gauge framework with $\mathbf{A}(t)$ naturally accommodates both electric and magnetic perturbations - Electric perturbation in the long-wavelength limit: $\mathbf{A}(t)$ spatially uniform, $\mathbf{B} = 0$ - Magnetic perturbation (symmetric gauge): $\mathbf{A}(\mathbf{r}) = \frac{1}{2}\mathbf{B}\times(\mathbf{r}-\mathbf{R}_G)$, spatially dependent, introduces gauge origin $\mathbf{R}_G$ - Note: magnetic perturbation sets up ECD/MCD as future extension (not implemented in present work but theoretically grounded here) - The gauge origin dependence with finite basis sets and the standard remedies (GIAO, IGLO) – brief mention, refer to Mattiat for details - Reference: Mattiat & Luber (2022) Section 2.3.2; Pickard & Mauri (2001)

[Note: This subsection establishes the theoretical foundation for magnetic perturbations. The current paper implements only electric perturbations (A(t) uniform). ECD/MCD calculations are future work. State this explicitly at the end of the subsection.]

2.5 Current Operator and Nonlocal Pseudopotential Contributions

This is the most important new subsection. Full derivation required. Content outline:

(a) Canonical vs mechanical momentum: - Canonical: $\hat{\mathbf{p}} = -\imath\nabla$ - Mechanical: $\hat{\boldsymbol{\pi}} = \hat{\mathbf{p}} + \mathbf{A}(t)$ - Velocity operator: $\hat{\mathbf{v}} = (1/\imath)[\hat{\mathbf{r}}, \hat{H}]$ - When $V^{\mathrm{nl}} \neq 0$: $\hat{\mathbf{v}} \neq \hat{\mathbf{p}}/m$ – extra commutator term $[\hat{\mathbf{r}}, V^{\mathrm{nl}}]$ - Ref: Starace (1971,1973); Mattiat & Luber eq.12

(b) Paramagnetic and diamagnetic current: - Full current: $\hat{\mathbf{J}} = \hat{\mathbf{J}}_{\mathrm{para}} + \hat{\mathbf{J}}_{\mathrm{dia}}$ - Paramagnetic: $\hat{\mathbf{J}}_{\mathrm{para}} \propto \mathrm{Re}[\psi^*(-\imath\nabla)\psi]$ - Diamagnetic: $\hat{\mathbf{J}}_{\mathrm{dia}} \propto \mathbf{A}(t)|\psi|^2$ - J_dia = 0 after t=0 in our perturbative scheme (A=0 for t>0) - Justify this explicitly as valid in linear response regime

(c) Continuity equation: - $\partial\rho/\partial t + \nabla\cdot\mathbf{J} = 0$ - Use as consistency/sanity check for the current operator

(d) Current in density matrix formulation: - Macroscopic current: $J_\alpha(t) = \mathrm{Re}\,\mathrm{Tr}[\mathbf{P}(t)\cdot\mathbf{J}^\alpha]$ - Current operator matrix: $J^\alpha_{ab} = p^\alpha_{ab} + [\hat{r}_\alpha, V^{\mathrm{nl}}]_{ab}$ - where $p^\alpha_{ab} = \langle\psi_a|(-\imath\nabla_\alpha + k_\alpha)|\psi_b\rangle$

(the k_α term comes from the Bloch theorem: see eq.XX) - This is what VecPHmatrix() + CurrentNlpp() compute in the code

(e) Nonlocal PP contribution $[r, V^nl]$: - For Kleinman-Bylander separable pseudopotentials: $V^{\text{nl}} = \sum_{I,lm} |\beta_I^{lm}\rangle h_{ij}^l \langle\beta_I^{lm}|$ - Compute $[\hat{r}_\alpha, V^{\text{nl}}]$ explicitly: $[\hat{r}_\alpha, V^{\text{nl}}]|\psi\rangle = \sum_{I,lm} |\beta_I^{lm}\rangle h_{ij}^l \langle[\hat{r}_\alpha, \beta_I^{lm}]|\psi\rangle$ - The commutator acts on the projector: $[\hat{r}_\alpha, \beta_I^{lm}]$ is computed by AppNls_0xyz() in the code - Extended algebra in Appendix A - Ref: Mattiat & Luber (2022) Section 2.2, eq.12; Pickard & Mauri (2001)

VERIFICATION NEEDED: trace the sign/i-factor from theory to code. VecPHmatrix() stores I_t * block_matrix (I_t = i) in Pxyz. CurrentNlpp() adds I_t * block_matrix. Current extracted as Re(zdotc(P, Px)). The -i from the gradient operator + i from storage = real observable. Verify this chain explicitly in the derivation. See Gap 1 in CONTEXT.md.

The velocity operator for a system described by a Hamiltonian containing a nonlocal pseudopotential V^{nl} is obtained from the Heisenberg equation of motion,

$$\hat{v}_\alpha = \frac{1}{\imath}[\hat{r}_\alpha, \hat{H}] = \hat{p}_\alpha + \frac{1}{\imath}[\hat{r}_\alpha, V^{\text{nl}}], \tag{30}$$

where the second term on the right-hand side arises because $[\hat{r}_\alpha, V^{\text{nl}}] \neq 0$ for a nonlocal operator.[?, ?] [Expand into full derivation of current operator, paramagnetic and diamagnetic decomposition, density matrix formulation, and explicit KB pseudopotential commutator. Follow derivation philosophy: complete first, then decide what moves to appendix.]

The macroscopic current density is expressed in terms of the one-particle density matrix $\mathbf{P}(t)$ as

$$J_\alpha(t) = \operatorname{Re}\operatorname{Tr}[\mathbf{P}(t) \cdot \mathbf{J}^\alpha]\,, \tag{31}$$

where $\mathbf{J}^\alpha$ is the matrix of current operator elements in the Kohn-Sham orbital basis,

$$J_{ab}^\alpha = \langle\psi_a| \left(-\imath\nabla_\alpha + k_\alpha + \frac{1}{\imath}[\hat{r}_\alpha, V^{\text{nl}}]\right) |\psi_b\rangle. \tag{32}$$

[The k_α term in eq. 32 arises from the Bloch theorem: for a periodic orbital $\psi_{n,\mathbf{k}}(\mathbf{r}) = e^{\imath\mathbf{k}\cdot\mathbf{r}}u_{n,\mathbf{k}}(\mathbf{r})$, the gradient operator acting on the full Bloch function gives $\nabla(e^{\imath\mathbf{k}\cdot\mathbf{r}}u) = e^{\imath\mathbf{k}\cdot\mathbf{r}}(\nabla + \imath\mathbf{k})u$. Derive this step explicitly.]

2.6 Density Matrix Propagation

[Adapt from paper 1 (Jakowski et al. 2025, Section 2.2).] The derivation is identical to paper 1 for the case S=1 (orthogonal basis at Gamma point). For $k \neq Gamma$, H is complex-valued (Hermitian but not real-symmetric) and Omega is also complex. The BCH expansion is the same but with complex matrix arithmetic. Key additions vs paper 1: - Note explicitly that for $k \neq Gamma$ the Hamiltonian matrix $\mathbf{H_k} = \langle\psi_a|\hat{H}|\psi_b\rangle$ is complex Hermitian - The BCH propagator (Ieldyn=1) handles complex matrices; the diagonalization path (Ieldyn=2) does NOT – state this - The self-consistency loop (predictor-corrector) is unchanged - The Hamiltonian at each time step includes the A(t)*p coupling term (for VECTOR_POT mode); in the perturbative kick scheme A appears only in the initial H_0 setup Reference sections from paper 1: Magnus expansion and density matrix propagation subsections. Cite: Jakowski & Morokuma (2009); Jakowski et al. JCTC 2025.

The time evolution of the one-particle density matrix $\mathbf{P}(t)$ is governed by the Liouville-von Neumann equation

$$\frac{\partial \mathbf{P}(t)}{\partial t} = -\imath \left[\mathbf{H}(t),\, \mathbf{P}(t)\right], \tag{33}$$

where $\mathbf{H}(t)$ is the Kohn-Sham Hamiltonian matrix in the basis of ground-state Kohn-Sham orbitals. [Continue with Magnus expansion (refer to paper 1 eqs. and cite), then commutator expansion for P(t+dt), self-consistency loop. For complex H: add note about BCH-only path.]

[Note from code inspection (CONTEXT.md Gap 4): magnus() in the code is documented as real-only, but for k≠ Γ the call site casts to double*. This is mathematically correct since the Magnus operation $\Omega = -\imath \cdot \frac{1}{2}(H_0 + H_1)\Delta t$ is element-wise and valid for both real and complex H, but should be stated explicitly in the text.]

2.7 Spin-Polarized Extension

Content outline: - Collinear spin-polarized DFT: two independent sets of KS orbitals $\{\psi_n^\uparrow\}$ and $\{\psi_n^\downarrow\}$, each with its own density matrix $\mathbf{P}^\uparrow(t)$ and $\mathbf{P}^\downarrow(t)$ - Spin-resolved electron density: $\rho^\uparrow(\mathbf{r},t)$ and $\rho^\downarrow(\mathbf{r},t)$ - Total density: $\rho = \rho^\uparrow + \rho^\downarrow$ - Spin magnetization: $m_s = \rho^\uparrow - \rho^\downarrow$ - Coupling between spin channels: through the spin-dependent XC potential $V_{\text{xc}}^\sigma[\rho^\uparrow, \rho^\downarrow]$ (LSDA or GGA spin-polarized functional) - Each spin channel propagates independently: $\partial\mathbf{P}^\sigma/\partial t = -\imath[\mathbf{H}^\sigma(t), \mathbf{P}^\sigma(t)]$ - The Hamiltonians $\mathbf{H}^\uparrow$ and $\mathbf{H}^\downarrow$ differ only through $V_{\text{xc}}^\uparrow \neq V_{\text{xc}}^\downarrow$ - Spin-resolved current and optical conductivity: $J_\alpha^\sigma(t) = \mathrm{Re}\,\mathrm{Tr}[\mathbf{P}^\sigma(t)\cdot\mathbf{J}^\alpha]\ \boldsymbol{\sigma}^\sigma(\omega)$ for each spin channel - Total and spin currents: $\mathbf{J}_{\text{charge}} = \mathbf{J}^\uparrow + \mathbf{J}^\downarrow\ \mathbf{J}_{\text{spin}} = \mathbf{J}^\uparrow - \mathbf{J}^\downarrow$ - Note: non-collinear spin (two-component spinors) is beyond the scope of this work and will be presented in a future publication. - Note from code inspection: spin is handled via separate MPI communicator groups (pct.spinpe); spin channels communicate only when updating V_{xc} (rho.get_oppo()).

In the collinear spin-polarized extension, the density matrix propagation (Section 2.6) is performed independently for each spin channel $\sigma \in \{\uparrow, \downarrow\}$,

$$\frac{\partial \mathbf{P}^\sigma(t)}{\partial t} = -\imath \left[\mathbf{H}^\sigma(t),\, \mathbf{P}^\sigma(t)\right], \tag{34}$$

where the spin-dependent Hamiltonian $\mathbf{H}^\sigma(t)$ contains the spin-resolved exchange-correlation potential $V_{\text{xc}}^\sigma[\rho^\uparrow(\mathbf{r},t), \rho^\downarrow(\mathbf{r},t)]$. [Continue: derive spin-resolved current, define total and spin currents, note that J_spin = J_up - J_down is available as a low-cost observable.]

2.8 Postprocessing: Optical Conductivity and Berry Phase

(a) From J(t) to optical conductivity: - Fourier transform: $\tilde{J}_\alpha(\omega) = \int J_\alpha(t) e^{\imath\omega t} dt$ - Optical conductivity tensor: $\sigma_{\alpha\beta}(\omega) = \tilde{J}_\alpha(\omega)/E_\beta(\omega)$ - Dielectric function: $\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + \imath\sigma_{\alpha\beta}(\omega)/\varepsilon_0\omega$ - In practice: apply delta-kick perturbation in direction β, Fourier transform $J_\alpha(t)$, divide by $E_\beta(\omega) = E_0$ (constant for delta kick) - Mention damping/broadening: apply $e^{-\gamma t}$ window before FFT

(b) Berry phase polarization as alternative observable: - For periodic systems, polarization can also be extracted via the King-Smith/Vanderbilt Berry phase formula:[?] $\mathbf{P}_{\mathrm{BP}} = \frac{e}{(2\pi)^3}\int_{\mathrm{BZ}} \mathbf{A}_n(\mathbf{k})\, d\mathbf{k}$ where $\mathbf{A}_n = \imath\langle u_{n\mathbf{k}}|\nabla_{\mathbf{k}}|u_{n\mathbf{k}}\rangle$ is the Berry connection - For the Gamma-point supercell limit, the discrete formula (Resta 1998, eq. 48 in Mattiat & Luber): $\langle\hat{d}_\alpha\rangle = \frac{eL}{2\pi}\mathrm{Im}\ln\det S_\alpha$ where $S_{\alpha,ij} = \langle\psi_i|e^{-2\pi\imath r_\alpha/L}|\psi_j\rangle$ - In density matrix form: $P_{\mathrm{BP},\alpha}(t) = \mathrm{Tr}[\mathbf{P}(t)\cdot\mathbf{X}^\alpha]$ where $\mathbf{X}^\alpha$ is the Berry phase position matrix - Equivalence with current-integration result: $\Delta P_\alpha = \int_0^t J_\alpha(t')\,dt'$ (test of internal consistency) - Implementation in RMG: BP_Xml matrix computed by Rmg_BP->tddft_Xml() (see code, RmgTddft.cpp lines 514,517) - Reference: King-Smith & Vanderbilt (1993); Resta (1998); Mattiat & Luber (2022) Section 2.5

The macroscopic current density $\mathbf{J}(t)$ obtained from eq. (31) provides direct access to the optical response of the periodic system. After applying a momentum kick of amplitude E_0 along the Cartesian direction β at $t = 0$, the optical conductivity tensor is obtained from the Fourier transform of the current density as

$$\sigma_{\alpha\beta}(\omega) = \frac{\tilde{J}_\alpha(\omega)}{E_\beta(\omega)} = \frac{1}{E_0}\int_0^\infty J_\alpha(t)\, e^{\imath\omega t-\gamma t}\, dt, \tag{35}$$

where γ is a broadening parameter.[?] [Expand: relate σ to ε; describe how three independent kick simulations (x, y, z) give the full tensor; note crystal symmetry reduces the number of independent components.]

[Add: Berry phase derivation and density matrix form. Show equivalence with current integration. Note that both observables are computed in the present implementation and their agreement serves as an internal consistency check.]

3 Implementation

The implementation of the periodic and spin-polarized RT-TDDFT presented in this work is built on the existing finite-system RT-TDDFT module of `RMG`,[?] which is described in detail in the accompanying paper. Here we focus on the extensions required for periodic boundary conditions and spin-polarized calculations, the key approximations and their physical justification, and the computational organization of the velocity gauge propagation.

3.1 Algorithm

The computational scheme for periodic RT-TDDFT in `RMG` is summarized in Algorithm 1. It follows the same density matrix propagation framework as the finite-system implementation[?] but with three key modifications: (1) the perturbation is applied as a momentum kick via the vector potential $\mathbf{A}$ rather than an electric field dipole kick; (2) the current operator matrix replaces the dipole matrix as the primary observable; and (3) the Hamiltonian matrix is evaluated at each k-point over the Brillouin zone.

Algorithm 1 Periodic RT-TDDFT Algorithm (VECTOR_POT mode)

/* **Ground state DFT calculations:** */
1: Generate Bloch orbitals $|\psi_{n,\mathbf{k}}\rangle$, Kohn-Sham Hamiltonian $\mathbf{H}^{\sigma}_{0,\mathbf{k}}$ for each k-point $\mathbf{k}$ and spin σ
2: Select active space: N_{occ} occupied + N_{virt} virtual orbitals

/* **Build current operator matrices (once, before time loop):** */
3: For each k-point $\mathbf{k}$ and spin σ:
4: Build paramagnetic current matrix:
$J^{\alpha}_{ab} \leftarrow \langle\psi_a|(-\imath\nabla_\alpha + k_\alpha)|\psi_b\rangle$ [VecPHmatrix()]
5: Add nonlocal PP correction:
$J^{\alpha}_{ab} \leftarrow J^{\alpha}_{ab} + \langle\psi_a|[\hat{r}_\alpha, V^{\mathrm{nl}}]|\psi_b\rangle$ [CurrentNlpp()]
6: Add $\mathbf{A}(t_0)\cdot\mathbf{J}$ to $\mathbf{H}_{-1,\mathbf{k}}$ to encode the perturbation kick at $t=0$

/* **Initialize density matrix:** */
7: $\mathbf{P}^0_{\mathbf{k}} \leftarrow \mathrm{diag}(f_{n,\mathbf{k}})$ (occupation numbers on diagonal)
8: $\mathbf{H}_{-1,\mathbf{k}} \leftarrow \mathbf{H}_{0,\mathbf{k}}$ (extrapolation seed)

/* **Time loop:** */
9: **for** $j = 1$ to N_{steps} **do**
10: $t \leftarrow t_0 + j\cdot\Delta t$
11: **Extrapolate:** $\mathbf{H}_{1,\mathbf{k}} \leftarrow 2\mathbf{H}_{0,\mathbf{k}} - \mathbf{H}_{-1,\mathbf{k}}$ [extrapolate_Hmatrix()]

/* **Self-consistent Hamiltonian loop:** */
12: **while** $\|\mathbf{H}^{\mathrm{new}}_1 - \mathbf{H}^{\mathrm{old}}_1\|_\infty > H_{\mathrm{thrs}}$ **do**
13: **Magnus operator:** $\boldsymbol{\Omega}_{\mathbf{k}} \leftarrow -\imath\cdot\frac{1}{2}(\mathbf{H}_{0,\mathbf{k}} + \mathbf{H}_{1,\mathbf{k}})\Delta t$ [magnus()]
14: **Propagate density matrix:** $\mathbf{P}^1_{\mathbf{k}} \leftarrow \mathbf{P}^0_{\mathbf{k}} + [\boldsymbol{\Omega}_{\mathbf{k}}, \mathbf{P}^0_{\mathbf{k}}] + \ldots$ [eldyn_ort() / commutp()]
15: **Update density:** $\rho^\sigma(\mathbf{r}) \leftarrow \rho^\sigma_{\mathrm{gs}}(\mathbf{r}) + \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_n P^1_{nn,\mathbf{k}}|\psi_{n,\mathbf{k}}(\mathbf{r})|^2$ [GetNewRho_rmgtddft()]
16: **Update potentials:** $V_{\mathrm{H}}[\rho]$, $V^{\sigma}_{\mathrm{xc}}[\rho^\uparrow,\rho^\downarrow]$
17: **Update Hamiltonian:** $\mathbf{H}^{\mathrm{new}}_{1,\mathbf{k}} \leftarrow \langle\psi_a|\hat{H}[\rho]|\psi_b\rangle$ [HmatrixUpdate()]
18: **end while**

19: **Extract current:** $J_\alpha(t) \leftarrow \mathrm{Re}\sum_{\mathbf{k}} w_{\mathbf{k}}\,\mathrm{Tr}[\mathbf{P}^1_{\mathbf{k}}\cdot\mathbf{J}^{\alpha}_{\mathbf{k}}]$
20: (Optional) **Berry phase polarization:** $P_{\mathrm{BP},\alpha}(t) \leftarrow \mathrm{Tr}[\mathbf{P}^1_{\mathbf{k}}\cdot\mathbf{X}^{\alpha}_{\mathbf{k}}]$
21: **Advance:** $\mathbf{P}^0 \leftarrow \mathbf{P}^1$; $\mathbf{H}_{-1} \leftarrow \mathbf{H}_0$; $\mathbf{H}_0 \leftarrow \mathbf{H}_1$
22: **end for**

/* **Postprocessing:** */
23: Fourier transform $J_\alpha(t)$ for each Cartesian direction α
24: Compute $\sigma_{\alpha\beta}(\omega)$, $\varepsilon_{\alpha\beta}(\omega)$

3.2 Key Approximations and Their Justification

The implementation described in Algorithm 1 embodies several approximations with respect to the full theory presented in Section 2. We discuss each in turn.

Perturbative momentum kick. The full velocity gauge Hamiltonian (eq. 29) contains the vector potential $\mathbf{A}(t)$ at every time step, requiring the $\mathbf{A}(t)\cdot\hat{\mathbf{p}}$ term to be re-evaluated throughout the propagation. In the present implementation, we instead apply $\mathbf{A}(t)$ as a delta-function kick at $t=0$ only: the momentum kick is encoded in the initial Hamiltonian $\mathbf{H}_{-1}$ by adding the $\mathbf{A}(t_0)\cdot\mathbf{J}^\alpha$ matrix (step 6 in Algorithm 1), after which $\mathbf{A}=0$ for all $t>0$. This approach is equivalent to first-order perturbation theory in $|\mathbf{A}_0|$, and it provides the full linear optical response of the system through the free oscillations of the current density $\mathbf{J}(t)$ following the kick. The linear response regime is valid as long as the kick amplitude is small enough that the system does not leave the linear regime, which is straightforwardly verified by checking that the response is proportional to $|\mathbf{A}_0|$.

Diamagnetic current term. The full current density contains both paramagnetic and diamagnetic contributions (Section 2.5). In the present implementation, the diamagnetic term $\mathbf{J}_{\mathrm{dia}} \propto \mathbf{A}(t)|\psi|^2$ is identically zero for all $t > 0$ because $\mathbf{A}(t > 0) = 0$ in the perturbative kick scheme. This is consistent with the linear response regime: the diamagnetic contribution to the linear response function is implicitly accounted for by the sum rule, and the paramagnetic current alone gives the correct optical conductivity.[?,?] Simulations with a continuously applied driving field (e.g., a CW laser or a Gaussian pulse) would require retaining $\mathbf{A}(t)$ throughout the propagation and are beyond the scope of the present work.

First-order Magnus expansion. As in the finite-system implementation,[?] the Magnus operator is truncated to first order:

$$\boldsymbol{\Omega} \approx \boldsymbol{\Omega}_1 = -\imath \cdot \tfrac{1}{2}(\mathbf{H}_0 + \mathbf{H}_1)\Delta t. \tag{36}$$

The second-order correction $\boldsymbol{\Omega}_2 = \frac{1}{12}\Delta t^2[\mathbf{H}_0, \mathbf{H}_1]$ is not included. The accuracy and stability of this scheme for the time steps used here was validated in paper 1 through energy conservation tests, and we demonstrate analogous stability for the periodic case in Section 4.

Norm-conserving pseudopotentials only. The current implementation supports only norm-conserving pseudopotentials (NCPP).[?,?] Ultrasoft pseudopotentials (USPP)[25] introduce a non-trivial overlap matrix $\mathbf{S} \neq \mathbf{I}$ into the density matrix propagation (the nonorthogonal basis path), which requires the generalized propagation scheme described in Appendix A of paper 1. This extension is planned for future work.

BCH propagator for complex Hamiltonians. For k-points away from the Γ point, the Hamiltonian matrix $\mathbf{H}_{\mathbf{k}}$ is complex Hermitian. In this case, the BCH commutator expansion propagator (Ieldyn=1) is used throughout; the diagonalization-based propagator (Ieldyn=2) is not currently supported for complex matrices. The BCH expansion is unconditionally stable and has been shown to give results equivalent to exact diagonalization within the convergence threshold,[?] and we find no practical limitation from this choice.

3.3 Software Optimization and Parallelization

[Content outline – adapt from paper 1 Section 3.2 with] additions for the periodic case: - Same MPI + OpenMP + GPU parallelization as paper 1 - k-point parallelization: each MPI group handles a subset of k-points (pct.kpsub_comm); current is summed over k-points with weights (MPI_Allreduce over kpsub_comm, lines 828-829 in RmgTddft.cpp) - Spin parallelization: separate MPI communicator groups per spin channel (pct.spin_comm); spin channels communicate only for XC update (MPI_Allreduce over spin_comm, line 767) - The density matrix P and Hamiltonian H are distributed using ScaLAPACK layout (Sp object); size is numst x numst where numst = N_occ + N_virt - For VECTOR_POT mode: momentum matrix J_alpha is complex even at Gamma point (stored as complex<double> in Pxmatrix_cpu etc.) - The BCH commutator $[\Omega, P]$ dominates $O(N^3)$; same optimization as paper 1 (real/imaginary decomposition, BLAS ZGEMM/DGEMM) - Timing data for benchmark systems: [add when available]

[Note: include a brief timing comparison table analogous to Table 1 in paper 1, but for one or more periodic benchmark systems (e.g., Si supercells of increasing size) to demonstrate scaling.]

4 Numerical Results

We benchmark the periodic and spin-polarized RT-TDDFT implementation in `RMG` against the `Octopus` code[16] for four representative systems of increasing dimensionality and physical complexity: a one-dimensional hydrogen chain (H_2), a two-dimensional hexagonal boron nitride (hBN) monolayer, a three-dimensional silicon crystal, and the spin-polarized magnetic insulator CrI_3. These systems are chosen to isolate and validate specific aspects of the implementation: k-point sampling, anisotropic optical response, three-dimensional periodic boundaries, and spin-collinear propagation, respectively. The section concludes with a demonstration of the unique large-scale capability of `RMG` through calculations on nitrogen-vacancy (NV) centers in diamond at varying supercell sizes.

In all calculations, a momentum kick of amplitude $E_0 =$ [value] a.u. was applied at $t = 0$ along each Cartesian direction independently, and the optical conductivity was obtained from the Fourier transform of the resulting current density. [Add: common computational parameters – time step, total propagation time, broadening gamma, XC functional, pseudopotential library used for all systems.]

4.1 One-Dimensional Periodic System: Hydrogen Chain

As the simplest possible periodic test system, we consider an infinite chain of hydrogen molecules aligned along the z axis with alternating bond lengths, modeled by a one-dimensional supercell with periodic boundary conditions. This system provides a stringent test of k-point sampling and the velocity gauge implementation in the limit of a quasi-one-dimensional geometry, where the longitudinal (z-direction) and transverse (x, y-direction) optical responses differ substantially.

[Computational details: - Supercell geometry: H–H bond length, H_2–H_2 separation - K-point sampling: [N_k points along z, Gamma in x and y] - Grid spacing: [value] a_0 - XC functional: [PBE or LDA] - Pseudopotential: [type] - Time step Δt: [value] a.u. - Total simulation time: [value] a.u. - Broadening γ: [value] eV]

[Discussion points: - Agreement between RMG and Octopus for both longitudinal and transverse components - K-point convergence: how many k-points needed for converged spectrum? - Physical interpretation of the spectral features - If Berry phase comparison is available: show agreement between current-integration and BP polarization as internal consistency check (Fruit 1 from CONTEXT.md) - If ground-state current convergence is shown: include as panel or note in text (Fruit 2 from CONTEXT.md)]

4.2 Two-Dimensional Periodic System: Hexagonal Boron Nitride

Hexagonal boron nitride (hBN) is a wide-gap insulating two-dimensional material with D_{6h} symmetry. Its optical response is highly anisotropic: the in-plane conductivity $\sigma_{xx}(\omega) =$

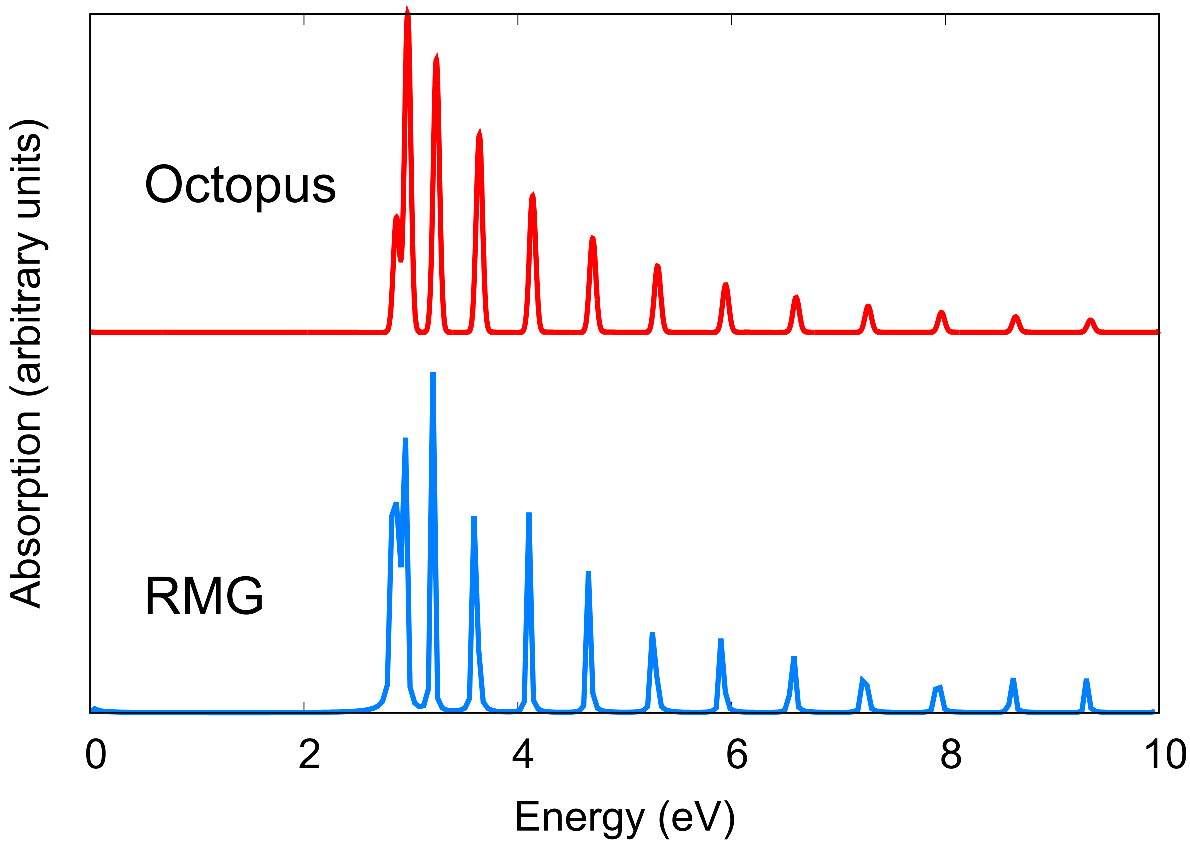

Figure 1: [Caption for H_2 chain optical conductivity figure. Should describe: (a) longitudinal $\sigma_{zz}(\omega)$ and transverse $\sigma_{xx}(\omega)$ components from RMG compared with Octopus; (b) convergence with k-point density if shown; labels for RMG and Octopus curves. Note anisotropy of 1D system as key feature to highlight.]

$\sigma_{yy}(\omega)$ differs markedly from the out-of-plane component $\sigma_{zz}(\omega)$, the latter being negligible for photon energies below the interlayer excitations. This system tests the implementation of two-dimensional periodic boundary conditions and the symmetry enforcement of the conductivity tensor in `RMG`.

[Computational details: - Monolayer geometry: lattice constant [a] Å, vacuum layer [d] Å to prevent inter-layer interactions - K-point sampling: [N_k × N_k × 1] Monkhorst-Pack - Grid spacing: [value] a_0 - XC functional: [PBE] - Pseudopotential: [type] - Time step and total simulation time: [values]]

[Discussion points: - Agreement between RMG and Octopus for in-plane conductivity - Symmetry: show that $\sigma_{xx} = \sigma_{yy}$ (Fruit 3 from CONTEXT.md) - Physical features: onset at band gap, first excitonic peak if visible - Comparison with available experimental or LR-TDDFT data if relevant]

4.3 Three-Dimensional Periodic System: Silicon

Silicon in the diamond cubic structure ($Fd\bar{3}m$, space group 227) is the canonical benchmark for periodic electronic structure calculations. Its indirect band gap and well-characterized

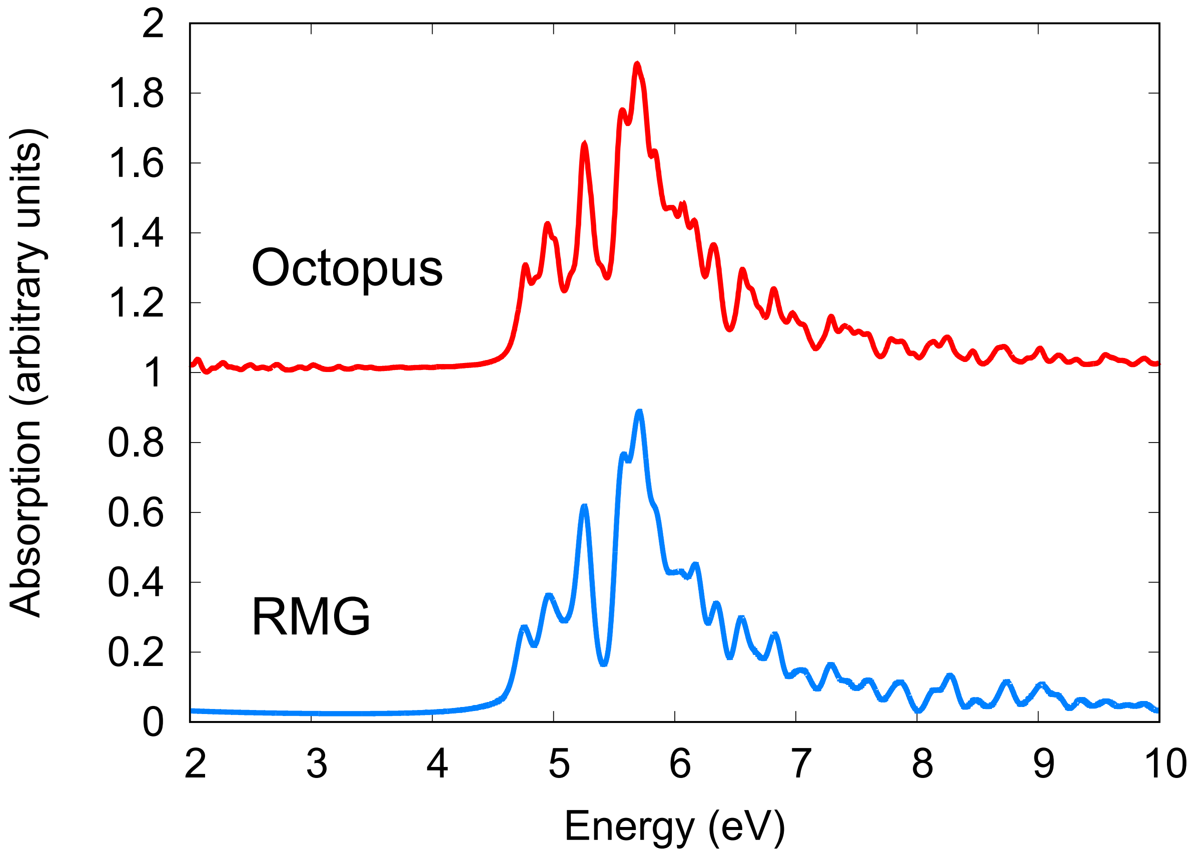

Figure 2: [Caption for hBN optical conductivity figure. Should describe: in-plane $\sigma_{xx}(\omega)$ from RMG and Octopus; out-of-plane component if shown; indicate the onset of absorption near the band gap. Highlight that $\sigma_{xx} = \sigma_{yy}$ (symmetry correctly reproduced) and $\sigma_{zz} \approx 0$.]

optical spectrum provide a rigorous test of the full three-dimensional periodic RT-TDDFT implementation. The cubic symmetry requires $\sigma_{xx}(\omega) = \sigma_{yy}(\omega) = \sigma_{zz}(\omega)$, providing an immediate symmetry validation.

[Computational details: - Unit cell: conventional 8-atom cell, lattice constant [a] Å- K-point sampling: [N_k × N_k × N_k] Monkhorst-Pack - Grid spacing: [value] a_0 - XC functional: [PBE] - Pseudopotential: [type, NCPP] - Number of occupied + virtual states: [N_occ + N_virt] - Time step Δt: [value] a.u. - Total simulation time: [value] a.u. ([value] fs) - Broadening γ: [value] eV]

[Discussion points: - Agreement between RMG and Octopus: peak positions, lineshapes - Symmetry: $\sigma_{xx} = \sigma_{yy} = \sigma_{zz}$ (Fruit 3) - Energy conservation test for periodic case (Fruit 5 from CONTEXT.md): run longer simulation and show no drift, analogous to paper 1 Fig. 3c - Berry phase comparison if included (Fruit 1) - Physical features: onset near direct gap, main optical peak structure]

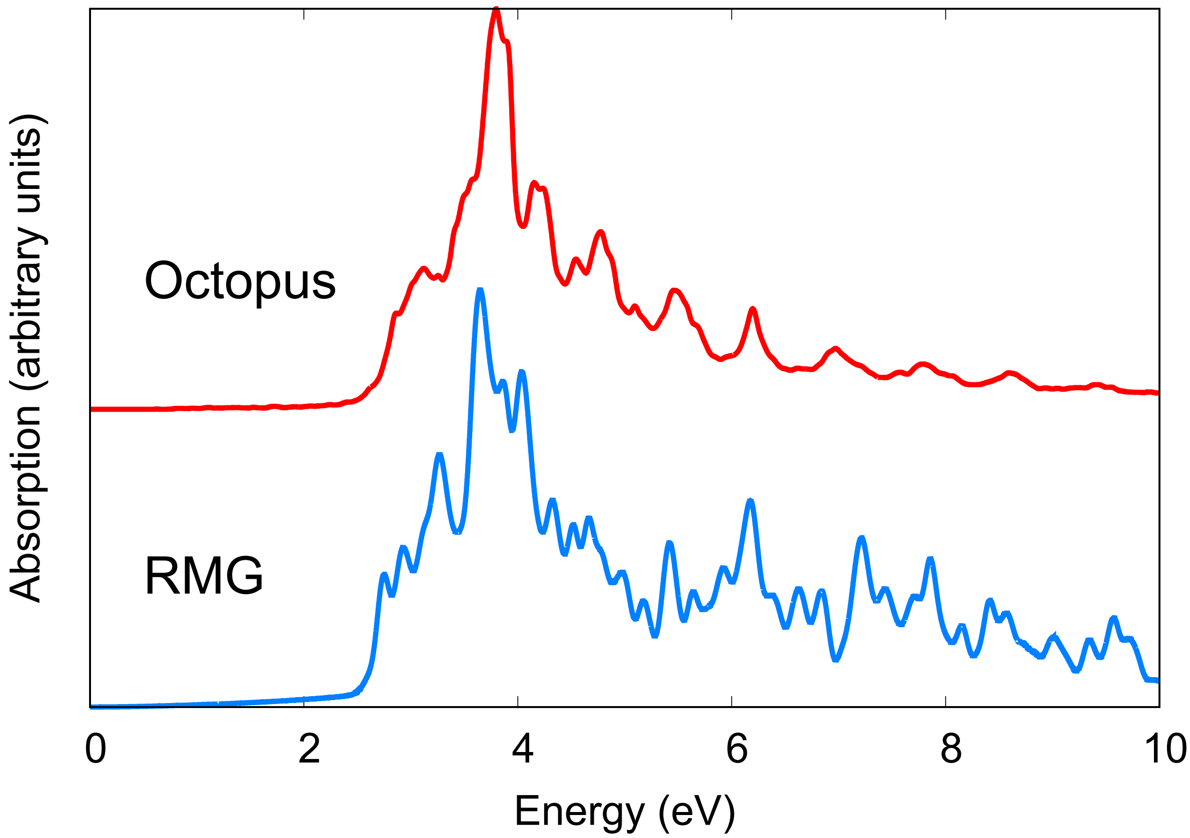

Figure 3: [Caption for Si optical conductivity figure. Should describe: $\sigma_{xx}(\omega)$ ($= \sigma_{yy} = \sigma_{zz}$ by cubic symmetry) from RMG compared with Octopus; indicate band gap region; optionally compare with experiment. If energy conservation panel included: describe it.]

4.4 Spin-Polarized System: CrI_3

Chromium triiodide (CrI_3) is a layered van der Waals magnetic insulator with a honeycomb lattice of Cr^{3+} ions, each carrying a local moment of approximately $3\,\mu_B$. In its ferromagnetic ground state, the spin-up and spin-down electronic structures differ, leading to distinct optical spectra for the two spin channels. This system provides a direct test of the spin-collinear RT-TDDFT extension: the two spin channels should propagate independently but with a common Hartree potential, yielding spin-resolved optical conductivities that can be compared with `Octopus` and with available experimental data.

[Computational details: - Unit cell: [describe geometry and magnetic ordering] - K-point sampling: [N_k × N_k × N_k] - XC functional: [PBE or PBE+U; specify U if used] - Pseudopotential: [type, NCPP] - Initial spin configuration: ferromagnetic, moments on Cr sites - Time step and total simulation time: [values] - Broadening: [value] eV]

[Discussion points: - Agreement between RMG and Octopus for both spin channels - Physical interpretation: which spectral features are spin-up vs spin-down? Charge-transfer excitations, Cr d-d transitions - Spin current $\mathbf{J}_{\text{spin}} = \mathbf{J}^{\uparrow} - \mathbf{J}^{\downarrow}$ if computed (Fruit 4 from CONTEXT.md) - Comparison with available experimental optical data if relevant - Note: ECD/MCD spectra not computed in this work (magnetic perturbation formalism reserved

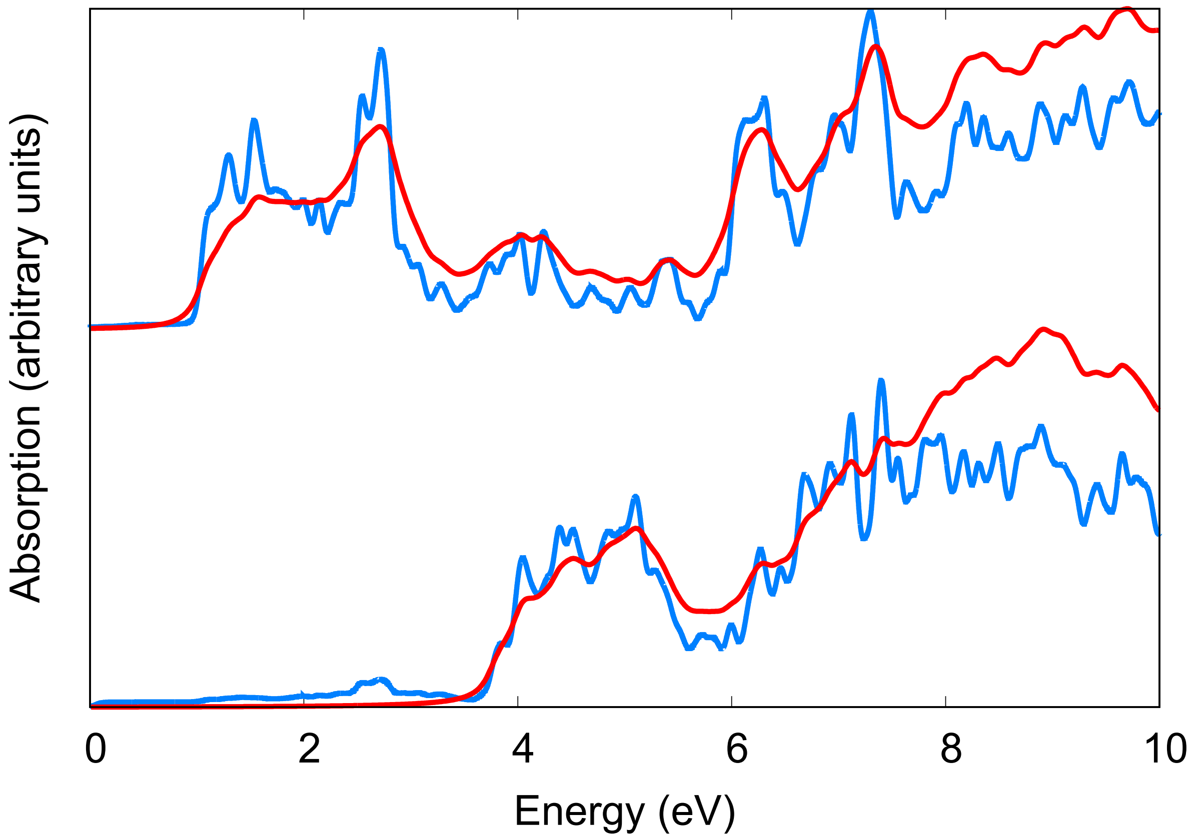

Figure 4: [Caption for CrI_3 optical conductivity figure. Should describe: spin-up and spin-down optical conductivities from RMG compared with Octopus; indicate the relevant spectral features (charge-transfer peaks, d-d transitions); note that spin channels are labeled consistently.]

for future publication)]

4.5 Large-Scale Demonstration: NV Center in Diamond

As a demonstration of the unique large-scale capability of `RMG`, we apply the periodic spin-polarized RT-TDDFT implementation to nitrogen-vacancy (NV) centers in diamond. The NV center—a substitutional nitrogen atom adjacent to a carbon vacancy—is one of the most studied point defects in solid-state physics, of primary importance for quantum sensing and quantum information applications due to its long spin coherence time at room temperature.[?]

In a standard periodic DFT or RT-TDDFT calculation, the NV center is represented by a single defect in a periodically repeated supercell. The physical properties of the *isolated* defect are recovered only when the supercell is large enough that the interaction between the defect and its periodic images is negligible. Identifying this convergence threshold is both a practical necessity and a physically meaningful result: it establishes the minimum defect concentration at which the isolated-defect limit is reached.

We investigate this convergence by performing RT-TDDFT calculations on diamond supercells containing a single NV center, systematically increasing the cell size from [N_{min}] to

[N_{max}] atoms. [Specific supercell sizes to be determined. Suggested sequence (subject to computational feasibility on Frontier): 64, 216, 512, 1000, [larger if possible] These correspond to defect concentrations of approximately [values] cm^{-3}.]

[Computational details: - Host: diamond cubic, lattice constant [a] Å- Defect: single NV^- center (negatively charged) - Spin state: triplet ground state ($S = 1$), ferromagnetic alignment - K-point sampling: Gamma point only for large supercells ([note if Gamma+additional k-points used for small cells]) - XC functional: [PBE] - Pseudopotential: [NCPP type] - Number of states: [N_occ + N_virt per spin] - Time step and total simulation time: [values] - Computational platform: Frontier at ORNL - Nodes used: [values per system size]]

[Figure placeholder: NV center convergence figure. Suggested panels: (a) Optical conductivity $\sigma(\omega)$ for each supercell size, showing convergence with increasing cell size (b) Convergence metric vs supercell size (e.g., peak position or integrated spectral weight) showing saturation (c) Optionally: timing data showing scaling with system size Label defect-related spectral features (zero-phonon line region, broad phonon sideband).]

Figure 5: [Caption for NV center convergence figure. Describe the system, the supercell sizes shown, the key spectral features, and the convergence behavior. Identify the minimum supercell size for which the isolated-defect optical spectrum is recovered within [tolerance].]

[Discussion points: - At what supercell size does the spectrum converge? What is the corresponding defect concentration threshold? - What is the physical origin of the convergence: direct defect-defect orbital overlap, or longer-range electrostatic interaction? - How does the spin-up and spin-down optical response differ? Does the convergence behavior differ between spin channels? - Timing data: how does the computational cost scale with cell size? At what size does the TDDFT step dominate over the SCF step? (Compare with paper 1 Table 1 format for Ag nanorods.) - This demonstration establishes the unique position of RMG for studying realistic defect concentrations in extended systems.]

4.6 Interaction of Matter with Electromagnetic Radiation

The starting point for any RT-TDDFT calculation is the coupling of the electronic system to an external time-dependent perturbation. This section establishes the general framework for the interaction of electrons with electromagnetic radiation, motivates the specific choices of gauge and approximation used in this work, and identifies the observables through which spectral properties are extracted from the simulation.

4.6.1 The general time-dependent Kohn-Sham equation with electromagnetic coupling

The starting point for RT-TDDFT in the presence of an electromagnetic field is the time-dependent Kohn-Sham (TDKS) equation with the standard minimal coupling prescription. For a single KS orbital ψ of an electron (charge $q_e = -e$, with $e > 0$), the TDKS equation reads

$$\imath\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \hat{H}(t)\, \psi(\mathbf{r}, t) \tag{37}$$

with the Hamiltonian

$$\hat{H}(t) = \frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{q_e}{c}\mathbf{A}(\mathbf{r},t)\right)^2 + q_e\,\Phi(\mathbf{r},t) + V_{\text{eff}}[\rho](\mathbf{r},t). \tag{38}$$

Here $\mathbf{A}(\mathbf{r},t)$ and $\Phi(\mathbf{r},t)$ are the vector and scalar potentials describing the electromagnetic field, and $V_{\text{eff}}[\rho]$ is the Kohn-Sham effective potential containing the ionic, Hartree, and exchange-correlation contributions.

In the general form, two electromagnetic potentials ($\mathbf{A}$ and Φ) and the KS potential all appear simultaneously. However, the gauge freedom of electrodynamics—the fact that $\mathbf{A}$ and Φ can be transformed without changing the physical fields $\mathbf{E}$ and $\mathbf{B}$—allows us to simplify significantly. The choice of gauge determines how the EM coupling enters the Hamiltonian.

Throughout this work we adopt the **Coulomb gauge**, defined by two conditions:

$$\nabla\cdot\mathbf{A} = 0 \quad \text{(transversality)}, \qquad \Phi_{\text{rad}} = 0 \quad \text{(no scalar potential from radiation)}. \tag{39}$$

In this gauge, all information about the radiation field is carried exclusively by $\mathbf{A}(\mathbf{r},t)$. The only scalar potential that remains is the instantaneous Coulomb interaction between charges, which is already contained in V_{eff}. The TDKS Hamiltonian simplifies to

$$\hat{H}(t) = \frac{1}{2m}\left(\hat{\mathbf{p}} + \frac{e}{c}\mathbf{A}(\mathbf{r},t)\right)^2 + V_{\text{eff}}[\rho](\mathbf{r},t), \tag{40}$$

where the sign flip ($-q_e \to +e$) comes from $q_e = -e$ for the electron.

Expanding the squared kinetic term gives three physically distinct contributions:

$$\hat{H}(t) = \underbrace{\frac{\hat{p}^2}{2m}}_{T_0} + \underbrace{\frac{e}{mc}\,\mathbf{A}\cdot\hat{\mathbf{p}}}_{\text{paramagnetic}} + \underbrace{\frac{e^2}{2mc^2}\,\mathbf{A}^2}_{\text{diamagnetic}} + V_{\text{eff}}. \tag{41}$$

The *paramagnetic term* is the leading-order light-matter coupling; it is linear in $\mathbf{A}$ and couples the field to the electronic current. The *diamagnetic term* is quadratic in $\mathbf{A}$ and is typically negligible for weak perturbative fields (it enters at second order in perturbation theory), though it is required for gauge invariance of the response and for satisfaction of the f-sum rule. In the Coulomb gauge, $\nabla\cdot\mathbf{A} = 0$ guarantees that $\mathbf{A}\cdot\hat{\mathbf{p}} = \hat{\mathbf{p}}\cdot\mathbf{A}$, so the paramagnetic term simplifies to a single $(e/mc)\mathbf{A}\cdot\hat{\mathbf{p}}$ without symmetrization.

[Molecular analogy:] In Gaussian basis set codes, the same Hamiltonian appears but $\hat{\mathbf{p}}$ is represented as matrix elements $\langle\phi_\mu|\hat{\mathbf{p}}|\phi_\nu\rangle$ between atomic orbitals. In the real-space grid representation used by RMG, $\hat{\mathbf{p}} = -\imath\hbar\nabla$ acts directly through finite-difference stencils on the grid—no basis transformation is needed.

4.6.2 The electromagnetic plane wave

A monochromatic EM plane wave in the Coulomb gauge is described by the vector potential

$$\mathbf{A}(\mathbf{r},t) = A_0\,\hat{\varepsilon}\,e^{\imath(\mathbf{q}\cdot\mathbf{r}-\omega t)} + \text{c.c.}, \tag{42}$$

where $\hat{\varepsilon}$ is the polarization unit vector, $\mathbf{q}$ is the photon wavevector with $|\mathbf{q}| = \omega/c$, and ω is the angular frequency. The transversality condition $\nabla \cdot \mathbf{A} = 0$ requires $\hat{\varepsilon} \cdot \mathbf{q} = 0$: the polarization is perpendicular to the direction of propagation.

The physical electric and magnetic fields are obtained from $\mathbf{A}$ alone through Maxwell's relations:

$$\mathbf{E}(\mathbf{r}, t) = -\frac{1}{c}\frac{\partial \mathbf{A}}{\partial t} = \frac{\imath \omega A_0}{c}\, \hat{\varepsilon}\, e^{\imath(\mathbf{q}\cdot\mathbf{r} - \omega t)} + \text{c.c.}, \tag{43}$$

$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A} = \imath A_0\, (\mathbf{q} \times \hat{\varepsilon})\, e^{\imath(\mathbf{q}\cdot\mathbf{r} - \omega t)} + \text{c.c.} \tag{44}$$

This confirms that a single vector potential in the Coulomb gauge is sufficient to represent both physical fields—no scalar potential is needed. In Gaussian units, $|\mathbf{B}|/|\mathbf{E}| = 1$ for a plane wave, but the coupling of non-relativistic electrons to $\mathbf{B}$ is weaker than to $\mathbf{E}$ by a factor of $v/c \approx \alpha \approx 1/137$. This hierarchy is the physical origin of the dominance of electric-dipole transitions in ordinary spectroscopy.

4.6.3 Multipole expansion and the long-wavelength approximation

The spatial factor $e^{\imath \mathbf{q}\cdot\mathbf{r}}$ in the vector potential can be expanded in a Taylor series:

$$e^{\imath \mathbf{q}\cdot\mathbf{r}} = 1 + \imath \mathbf{q} \cdot \mathbf{r} - \tfrac{1}{2}(\mathbf{q} \cdot \mathbf{r})^2 + \cdots \tag{45}$$

Each term corresponds to a class of multipole transitions with distinct selection rules and characteristic coupling strengths.

The **zeroth-order term** (1) retains only the temporal oscillation of $\mathbf{A}$, making it spatially uniform: $\mathbf{A}(\mathbf{r}, t) \to \mathbf{A}(t)$. This is the **long-wavelength** or **dipole approximation**. The resulting $(e/mc)\mathbf{A}(t) \cdot \hat{\mathbf{p}}$ interaction drives electric dipole (E1) transitions—the dominant contribution to optical absorption. The approximation is valid when the spatial extent of the electronic system is much smaller than the photon wavelength: $|\mathbf{q}||\mathbf{r}| \ll 1$. For visible light ($\lambda \approx 400$–$700$ nm) interacting with molecules ($\sim$1 nm) or even nanoscale systems ($\sim$10 nm), the ratio $|\mathbf{q}||\mathbf{r}| \approx 10^{-2}$–$10^{-1}$ is small and the approximation is excellent.

The **first-order term** ($\imath \mathbf{q} \cdot \mathbf{r}$) gives the electric quadrupole (E2) and magnetic dipole (M1) contributions. These are suppressed by a factor of $|\mathbf{q}||\mathbf{r}| \approx v/c$ relative to E1 and are important for forbidden transitions, circular dichroism, and X-ray spectroscopy where λ approaches atomic dimensions.

Higher-order terms give electric octupole, magnetic quadrupole, etc., relevant only for very short wavelengths or very extended systems.

The long-wavelength approximation (retaining only the zeroth-order term) has been the standard in molecular quantum chemistry, including in paper 1 of this series.[?] However, it has well-known limitations that become critical for periodic systems, as discussed next.

4.6.4 Failure of the length gauge for periodic systems

In the long-wavelength limit with spatially uniform $\mathbf{A}(t)$, two equivalent representations of the light-matter interaction are available, connected by the Göppert-Mayer gauge transformation (derived in Section **??**):

Velocity gauge (VG). The interaction enters through the kinetic energy as $(e/mc)\mathbf{A}(t)\cdot\hat{\mathbf{p}}$. The perturbation operator involves $\hat{\mathbf{p}} = -\imath\hbar\nabla$, which is well-defined for any basis set or boundary condition.

Length gauge (LG). A gauge transformation with $\chi(\mathbf{r}, t) = -\mathbf{r}\cdot\mathbf{A}(t)$ eliminates $\mathbf{A}$ from the kinetic energy and transfers the interaction to a scalar potential $\Phi = \mathbf{E}(t)\cdot\mathbf{r}$, giving the familiar dipole interaction $-e\,\mathbf{E}(t)\cdot\hat{\mathbf{r}}$. The perturbation operator now involves the position operator $\hat{\mathbf{r}}$.

Note that these two gauges both operate within the Coulomb gauge framework—they are different representations of the same physics obtained by choosing different gauge functions, not different physical theories. For *finite* molecular systems, both give identical results, and the length gauge is often preferred because the dipole matrix elements $\langle\psi_a|\hat{r}_\alpha|\psi_b\rangle$ have a transparent physical interpretation as transition dipole moments. This was the approach used in paper 1.[?]

For *periodic* systems under Born-von Kármán boundary conditions, the length gauge fails catastrophically. The position operator $\hat{\mathbf{r}}$ is fundamentally incompatible with periodicity: $\mathbf{r}$ and $\mathbf{r}+\mathbf{L}$ must be physically equivalent, but $\hat{\mathbf{r}}$ distinguishes them. The scalar potential $\mathbf{E}\cdot\mathbf{r}$ is a linearly growing ramp that produces a discontinuous "sawtooth" at the unit cell boundary—it is not periodic and cannot be represented in a Bloch basis. The matrix elements $\langle\psi_{n\mathbf{k}}|\hat{r}_\alpha|\psi_{m\mathbf{k}}\rangle$ between Bloch states are ill-defined (this is precisely the problem resolved by the modern theory of polarization via Berry phases, discussed in Section 4.6.7).

The velocity gauge completely avoids this problem: the momentum operator $\hat{\mathbf{p}}$ is perfectly well-defined for Bloch states, and the matrix elements $\langle\psi_{n\mathbf{k}}|\hat{p}_\alpha|\psi_{m\mathbf{k}}\rangle$ exist without any difficulty. This is why periodic-system RT-TDDFT implementations, including the present one, use the velocity gauge.

4.6.5 Scope and limitations of the velocity gauge with the dipole approximation

The velocity gauge resolves the periodic-boundary problem, but the underlying long-wavelength approximation itself imposes a fundamental limitation: it restricts the calculation to **vertical transitions** with zero crystal momentum transfer ($\Delta\mathbf{k} = 0$). This means that only the optical limit of the dielectric function $\varepsilon(\mathbf{q}\to 0, \omega)$ is accessible.

It is sometimes stated that the velocity gauge "is not general," but this is imprecise. The gauge transformation itself is exact—gauge transformations are always unitary and introduce no approximation. What is limited is the *dipole approximation* that precedes it: by setting $e^{\imath\mathbf{q}\cdot\mathbf{r}} \to 1$, one discards all finite-$\mathbf{q}$ physics regardless of which gauge is used. The velocity gauge is the correct and rigorous framework for $q = 0$ optical response in periodic systems; the limitation lies in the $q = 0$ restriction, not in the gauge.

Many important spectroscopies and collective excitations involve finite momentum transfer and lie beyond this approximation. These include plasmon dispersion $\varepsilon(\mathbf{q}, \omega)$, electron energy-loss spectroscopy (EELS) where the beam transfers finite $\mathbf{q}$ to the sample, inelastic X-ray scattering (IXS), and magnon dispersion in magnetic systems.

4.6.6 Beyond the dipole approximation: the full vector potential

A natural generalization is to retain the full spatial dependence of the vector potential without making the dipole approximation:

$$\hat{H}(t) = \frac{1}{2m}\Big(\hat{\mathbf{p}} + \frac{e}{c}\mathbf{A}(\mathbf{r},t)\Big)^2 + V_{\text{eff}}[\rho](\mathbf{r},t) \tag{46}$$

with $\mathbf{A}(\mathbf{r},t) = A_0\hat{\varepsilon}\, e^{\imath(\mathbf{q}\cdot\mathbf{r}-\omega t)} + \text{c.c.}$ retaining the full plane-wave spatial structure. This is still the Coulomb gauge—no new gauge choice is needed. The spatial factor $e^{\imath\mathbf{q}\cdot\mathbf{r}}$ is automatically periodic (it is a reciprocal-space function commensurate with the lattice for appropriate $\mathbf{q}$), so the problematic $\hat{\mathbf{r}}$ operator never appears. Transitions now satisfy crystal momentum conservation $\mathbf{k}_f = \mathbf{k}_i + \mathbf{q}$ (mod $\mathbf{G}$), allowing finite-$\mathbf{q}$ excitations.

This approach is more general than the dipole approximation because the full exponential automatically includes all multipole orders. For finite but small $|\mathbf{q}||\mathbf{r}|$, the leading correction beyond E1 is the electric quadrupole / magnetic dipole term (first order in $\mathbf{q}\cdot\mathbf{r}$). For $|\mathbf{q}||\mathbf{r}| \sim 1$ (as in X-ray spectroscopy or large nanostructures), all orders contribute and the full exponential must be retained.

Important distinction regarding finite-q response. There is a second, independent route to finite-$\mathbf{q}$ excitations that does not involve the vector potential at all. One can apply a scalar potential perturbation

$$\delta V(\mathbf{r},t) = V_0\, e^{\imath\mathbf{q}\cdot\mathbf{r}}\, \delta(t) \tag{47}$$

directly to V_{eff}, with no gauge considerations. This "density-wave kick" probes the density-density response function $\chi(\mathbf{q},\omega)$, whereas the EM route probes the current-current response $\sigma(\mathbf{q},\omega)$. The two response functions are related but distinct: σ is a tensor (3×3), χ is a scalar; σ naturally gives the transverse (optical) response, χ gives the longitudinal response relevant to EELS. For the longitudinal response at finite $\mathbf{q}$, the scalar perturbation approach is more direct. For transverse (optical) response including finite-$\mathbf{q}$ corrections (beyond dipole), the full $\mathbf{A}(\mathbf{r},t)$ approach is the natural choice.

The present implementation uses the long-wavelength limit (spatially uniform $\mathbf{A}(t)$) exclusively. The general framework described here indicates the natural extension path: keeping the full $e^{\imath\mathbf{q}\cdot\mathbf{r}}$ spatial structure in $\mathbf{A}$ for beyond-dipole optical response, or using scalar density-wave perturbations for longitudinal excitations at arbitrary $\mathbf{q}$. Such extensions would enable simulation of plasmon dispersion, EELS spectra, inelastic X-ray scattering, and magnon dispersion from first principles—all within the same RT-TDDFT framework.

4.6.7 Observables: from dipole moment to current density to Berry phase polarization

The choice of gauge determines not only how the perturbation enters the Hamiltonian but also which observable provides the most natural route to spectral properties. Three complementary approaches exist, corresponding to three equivalent but operationally distinct ways of characterizing the electronic response.

Length gauge—dipole moment. In the length gauge, the EM field couples through the scalar potential $\mathbf{E}(t) \cdot \hat{\mathbf{r}}$, acting directly on the charge density. The natural observable is the time-dependent dipole moment:

$$\mu_\alpha(t) = -e \sum_n f_n \langle \psi_n(t) | \hat{r}_\alpha | \psi_n(t) \rangle, \tag{48}$$

where the sum runs over occupied KS orbitals with occupations f_n. The Fourier transform of $\mu(t)$ after an impulsive perturbation yields the frequency-dependent polarizability tensor $\alpha(\omega)$, from which the photoabsorption cross section follows as $\sigma_{\text{abs}}(\omega) = (4\pi\omega/c)\,\text{Im}\,\alpha(\omega)$. For finite molecular systems this is the standard RT-TDDFT post-processing route, and it was used in paper 1.[?]

Higher-order multipole moments—the quadrupole $Q_{\alpha\beta}(t) = \langle r_\alpha r_\beta \rangle$, the octupole, etc.—can be extracted analogously and provide access to E2, M1, and higher-order contributions to optical activity and circular dichroism, though these are rarely needed for systems small compared to the photon wavelength.

Velocity gauge—electronic current density. In the velocity gauge, the EM field couples through $\mathbf{A}(t) \cdot \hat{\mathbf{p}}$, acting on the electronic momentum. The natural observable is the time-dependent electronic current density:

$$\mathbf{j}(\mathbf{r}, t) = \frac{1}{2} \sum_n f_n \left[\psi_n^*(\mathbf{r}, t)\, \hat{\mathbf{p}}\, \psi_n(\mathbf{r}, t) + \text{c.c.} \right] + \mathbf{j}_{\text{NL}}(\mathbf{r}, t). \tag{49}$$

The first term is the familiar paramagnetic current—the same quantity that appears in probability current expressions in introductory quantum mechanics. The second term, $\mathbf{j}_{\text{NL}}$, arises from the nonlocal pseudopotential: because the Kleinman-Bylander projectors do not commute with the position operator, the continuity equation $\partial\rho/\partial t + \nabla \cdot \mathbf{j} = 0$ is satisfied only when the nonlocal contribution is included. The derivation and explicit form of $\mathbf{j}_{\text{NL}}$ are given in Section 2.5. [Verify cross-reference label for Sec. 2.5.]

The macroscopic (cell-averaged) current $\mathbf{J}(t) = (1/\Omega) \int_{\text{cell}} \mathbf{j}(\mathbf{r}, t)\, d^3r$ connects to optical properties through the frequency-dependent optical conductivity:

$$\sigma_{\alpha\beta}(\omega) = \frac{J_\alpha(\omega)}{E_\beta(\omega)}, \tag{50}$$

and the dielectric function:

$$\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + \frac{4\pi\imath}{\omega}\, \sigma_{\alpha\beta}(\omega), \tag{51}$$

from which absorption spectra, reflectivity, refractive index, and energy-loss functions follow by standard relations. This is the periodic-system analog of the molecular polarizability route: where molecular RT-TDDFT extracts $\alpha(\omega)$ from $\mu(t)$, periodic RT-TDDFT extracts $\varepsilon(\omega)$ from $\mathbf{J}(t)$.

Berry phase—macroscopic polarization. The macroscopic polarization $\mathbf{P}$ of a periodic system cannot be defined as the expectation value of any local operator—this is the same fundamental difficulty with $\hat{\mathbf{r}}$ discussed in Section 4.6.4. The modern theory of polarization? ? resolves this by expressing $\mathbf{P}$ as a geometric phase (Berry phase) of the occupied Bloch orbitals:

$$P_\alpha = \frac{-e}{(2\pi)^3} \sum_n \int_{\mathrm{BZ}} d\mathbf{k}\, \langle u_{n\mathbf{k}} | \imath \frac{\partial}{\partial k_\alpha} | u_{n\mathbf{k}} \rangle, \tag{52}$$

where $u_{n\mathbf{k}}(\mathbf{r})$ is the cell-periodic part of the Bloch orbital. This expression is defined modulo a quantum of polarization $e\mathbf{R}/\Omega$ (where $\mathbf{R}$ is a lattice vector and Ω is the cell volume), reflecting the intrinsic multivaluedness of the geometric phase.

The Berry phase polarization is the periodic-system generalization of the molecular dipole moment. Just as $\mu = -e\langle\hat{\mathbf{r}}\rangle$ is a single well-defined number for a localized molecule, P^{Berry} is a well-defined number (modulo the quantum) for an infinite periodic crystal. The Berry phase accomplishes for periodic systems what the position operator does for molecules—but through a topological construction in k-space rather than a real-space expectation value.

In RT-TDDFT, the time-evolved Bloch orbitals $\psi_{n\mathbf{k}}(t)$ at each time step define a time-dependent Berry phase, giving the polarization $\mathbf{P}(t)$ directly without reference to the position operator. The connection to the current is exact:

$$\mathbf{J}(t) = \frac{d\mathbf{P}}{dt}, \tag{53}$$

so the polarization can equivalently be recovered by time-integration of the current:

$$\mathbf{P}(t) = \mathbf{P}(0) + \int_0^t \mathbf{J}(t')\, dt'. \tag{54}$$

The two routes—direct Berry phase evaluation at each step versus current integration—are formally equivalent but have different numerical characteristics. The Berry phase gives $\mathbf{P}(t)$ without accumulating integration error, but requires dense k-point sampling (the formula is a k-space integral) and careful tracking of the polarization quantum for large-amplitude perturbations. Current integration avoids the Berry phase calculation but accumulates time-stepping error over long propagations. In the linear response regime (small perturbations), both approaches are reliable and serve as mutual consistency checks. The Berry phase formalism and its implementation in the present code are described in detail in Section **??**. [Verify cross-reference label for Sec. 2.8.]

Equivalence, complementarity, and the choice of observable. For finite systems where $\hat{\mathbf{r}}$, $\hat{\mathbf{p}}$, and the Berry phase are all well-defined, the three routes are exactly equivalent, connected by the time derivative of the dipole moment:

$$\frac{d\mu_\alpha}{dt} = -e \sum_n f_n \langle \psi_n(t) | \hat{v}_\alpha | \psi_n(t) \rangle = \Omega\, J_\alpha(t) = \Omega\, \frac{dP_\alpha}{dt}. \tag{55}$$

In frequency space this yields the fundamental relation $\sigma(\omega) = \imath\omega\,\alpha(\omega)$, linking the polarizability to the conductivity, and the dielectric function $\varepsilon(\omega) = 1+4\pi\alpha(\omega)/\Omega = 1+4\pi\imath\sigma(\omega)/\omega$.

For periodic systems, the dipole moment is undefined, leaving two viable routes: the current density $\mathbf{J}(t)$, which gives $\sigma(\omega)$ and $\varepsilon(\omega)$ directly via Fourier transform; and the Berry phase $\mathbf{P}(t)$, which gives $\alpha(\omega)$ and connects to the absolute polarization. In the present work, we use the current density as the primary observable (Sections 2.5–[2.6]) and the Berry phase as a complementary diagnostic and for properties where the absolute polarization is needed (Section **??**).

There is an appealing conceptual symmetry in the gauge-observable pairing: in the length gauge, the field perturbs the density (through $\mathbf{E} \cdot \hat{\mathbf{r}}$) and the response is read from the density's moments; in the velocity gauge, the field perturbs the current (through $\mathbf{A} \cdot \hat{\mathbf{p}}$) and the response is read from the current. The Berry phase stands apart from this pattern—it is a gauge-invariant geometric property of the occupied manifold that can be evaluated regardless of how the perturbation was applied, depending only on the instantaneous state of the system, not on the history of how it was driven.

4.6.8 Summary and scope of the present work

The general TDKS equation with full minimal coupling to $\mathbf{A}(\mathbf{r}, t)$ in the Coulomb gauge provides a unified framework for modeling the interaction of electronic systems with electromagnetic radiation. The key approximations adopted in the present work and their consequences are as follows.

The **Coulomb gauge** ($\nabla \cdot \mathbf{A} = 0$, $\Phi_{\text{rad}} = 0$) carries all radiation physics in $\mathbf{A}$ alone, avoiding ambiguity between scalar and vector potential contributions. Both electric and magnetic fields are derivable from $\mathbf{A}$ via standard Maxwell relations.

The **long-wavelength approximation** ($e^{i\mathbf{q}\cdot\mathbf{r}} \to 1$) reduces $\mathbf{A}(\mathbf{r}, t)$ to a spatially uniform $\mathbf{A}(t)$, restricting the calculation to vertical transitions ($\Delta\mathbf{k} = 0$). This is equivalent to the electric dipole approximation and is valid when the photon wavelength far exceeds the system size.

The **velocity gauge** represents the interaction as $(e/mc)\mathbf{A}(t) \cdot \hat{\mathbf{p}}$ in the kinetic energy, avoiding the ill-defined position operator that plagues the length gauge in periodic systems.

The **current density** is the primary observable, giving the optical conductivity $\sigma(\omega)$ and dielectric function $\varepsilon(\omega)$ via Fourier transform. The **Berry phase polarization** provides a complementary route to the same spectral information and additionally gives the absolute macroscopic polarization.

The present work operates within all three approximations: Coulomb gauge, long-wavelength limit, velocity gauge. This gives access to the full optical response $\varepsilon(\omega)$ of periodic systems with correct treatment of local field effects, current contributions from nonlocal pseudopotentials, and spin-collinear dynamics. Extensions to finite-$\mathbf{q}$ response—via either the full $\mathbf{A}(\mathbf{r}, t)$ or scalar density-wave perturbations—and to higher-multipole processes are natural next steps that the framework accommodates without conceptual difficulty.

5 Summary and Outlook

[To be written after all other sections are complete. Should summarize the key contributions, state the validated capabilities, and outline planned extensions: (1) continuous driving field / nonlinear response, (2) non-collinear spin RT-TDDFT, (3) Ehrenfest dynamics combined with spin, (4) ultrasoft pseudopotentials.]

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A Gauge Transformation of the Nonlocal Pseudopotential Term

[Full derivation of the gauge transformation of Vˆnl from length to velocity gauge, following the structure of Mattiat & Luber (2022) Appendix B but adapted to the Kleinman-Bylander form used in RMG. This is the extended algebra supporting section 2 subsection on current operator and nonlocal pseudopotentials.]

B Equivalence of Current-Based and Berry Phase Polarization

[Proof that the velocity gauge current J(t), when integrated, gives the same polarization as the King-Smith/Vanderbilt Berry phase formula. This supports the comparison presented in section **??**.]

C [Additional appendix if needed]

[Placeholder. May be used for: spin-collinear extension of the BCH scheme; extended orthogonalization proofs if needed beyond paper 1 appendix; or can be removed if not needed.]

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